





Bachelor Thesis

Quantization and Spike-Based Explainability in Spiking Neural Networks for Gravitational-Wave Glitch Classification

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Abstract

Quantization makes a neural network cheap enough to deploy, and it is usually validated by checking that accuracy does not fall. On convolutional networks, accuracy has been shown to survive quantization while the explanations of the model's decisions do not. This thesis asks the spiking analogue: *does a quantized spiking neural network base its decisions on the same evidence as its full-precision counterpart, and at what energy cost?* A spiking network carries a membrane potential that can be quantized independently of the weights, and its explanations are sequences of maps; the working hypothesis is that quantizing the membrane degrades the temporal structure of an explanation before quantizing the weights degrades its spatial structure.

A four-block convolutional spiking network is trained on four classes of transient noise from the LIGO Gravity Spy release, reaching 0.9833 ± 0.0014 validation accuracy over three seeds under hardware-realizable constraints, and its decisions are explained with the gradient-free Spike Activation Map. Before any comparison, the explanation metrics are calibrated: two full-precision models differing only in seed produce maps correlating at 0.153 on the worst class, against a numerical floor of 0.975. Spatial metrics are therefore valid only between a model and its own parent, and every quantized model is fine-tuned from, and compared with, the full-precision model of its own seed.

Weights only, membrane only, and both are quantized at 8, 4 and 2 bits. At 8 and 4 bits accuracy and explanations stay at the level of the parent in every arm. At 2 bits the arms separate: ternary weights collapse the classifier, while a 2-bit membrane keeps accuracy at parity (0.982) and yet changes the explanation, whose highlighted region overlaps the parent's by 0.44 against a floor of 0.70 — a change the spatial correlation (0.92) would have reported as a preserved explanation. The new maps remain faithful by the deletion metric: they point at evidence the model uses, but not at the evidence the original pointed at. The hypothesis is not supported as an ordering of explanation metrics, but its mechanism is: the 2-bit membrane makes the spike trains of the first layer perfectly periodic, a temporal change the explanation's temporal metric cannot see. That configuration is also dominated on energy and memory, so there is no trade-off between efficiency and explainability here: a joint 4-bit model keeps both at about one sixtieth of the estimated energy and one eighth of the memory of its parent.

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Chapter 1
Introduction

1.1 Motivation

In September 2015 the two Advanced LIGO (Laser Interferometer Gravitational-Wave Observatory) detectors made the first direct detection of a gravitational wave, GW150914, which was produced by the merger of two black holes [1]. Two years later, LIGO and Virgo jointly observed GW170817, the first signal from a binary neutron star merger [2]. Its electromagnetic counterpart, from a short gamma-ray burst to a kilonova, was then followed by telescopes around the world, an event that marked the beginning of multi-messenger astronomy [3].

Since then, detections have gone from exceptional events to routine ones, leading to a continuous stream of data in the field of gravitational-wave astronomy, data that must be analyzed, often in real time. In this setting, the main obstacle is not the faintness of the astrophysical signals but the noise. The detectors are sensitive enough to record a constant stream of short, non-Gaussian transients of instrumental and environmental origin, known as *glitches*. What makes glitches both interesting and detrimental to astronomical analysis is that they can mimic genuine signals or overlap with them, which raises the false-alarm rate and reduces the effective sensitivity of searches.

Gravity Spy, the source of the data used in this work, is an Open Source project specifically created to address this problem. It combines citizen-science classification with machine learning, aiming to sort glitches into morphological families based on their time–frequency spectrograms [44]. Knowing which family a glitch belongs to is the first step towards tracing it back to its physical cause and, where possible, addressing the source of the problem and ultimately removing that cause at the instrument.

More broadly, machine learning has become indispensable to perform data-quality work, but it comes at a cost. The demand for computing power is indeed growing faster than the resources available, public alerts must be issued within minutes of a detection, and data volumes are expected to grow further with future observing runs and next-generation detectors. Making models lighter and cheaper to run is therefore a natural goal. Two approaches are particularly relevant: quantization, which compresses an existing network, and spiking neural networks (SNNs), which are designed from the outset for energy-efficient, and event-driven computation.

Running a neural network outside a data center generally means making it smaller, and the standard tool for this is indeed quantization. Quantization is a practice implemented to represent a network's weights and activations with fewer bits, trading numerical precision for savings in memory and energy. A quantized model is routinely validated by checking that its accuracy has not dropped, and for many applications this is a sufficient test.

It is not sufficient, however, when the model is also expected to explain its decisions. Rabbi et al. [33] evaluated five convolutional architectures at INT8 and INT4, comparing the attribution maps of the quantized models with those of their full-precision counterparts. Their core finding was that explanation similarity degrades measurably between

INT8 and INT4, even where accuracy differences are negligible. They also discovered that when quantization does cause a misclassification, the quantized model is generally attending to *different* regions of the input, rather than weighing the same evidence differently.

In other words, accuracy and interpretability do not degrade together.

This finding, however, is limited to conventional artificial neural networks (ANNs), and spiking neural networks (SNNs), the architecture most closely associated with low-power deployment [36], differ from ANNs in ways that make the question both harder and more interesting.

1.2 Research question

The question this thesis addresses is:

> *Does a quantized spiking neural network under edge-realistic constraints base its decisions on the same evidence as its full-precision counterpart, and at what energy cost?*

Spiking networks possess three properties that are absent in convolutional networks, and each of them is crucial for framing the research question.

A second quantization target. A spiking neuron carries a membrane potential which is a state variable read and written at every time step, and whose storage scales with the number of neurons, rather than the number of parameters. In the network studied here it occupies 367.5 KiB against 1 152.7 KiB of weights at full precision, and it becomes the larger share of the footprint as soon as the weights are compressed. It is a quantization target of its own [43], and it can and will be quantized independently of the weights [38], whereas a conventional network offers only weights and activations.

A temporal axis in the explanation. A spiking explanation is a *sequence* of maps, one per time step, which effectively builds a temporal axis in the explanations. Quantization may preserve where a network looks while displacing when it decides, a change to which spatial metrics are blind by construction.

A native explainer. The Spike Activation Map (SAM) [22] reads spike activity directly. It requires neither gradients, which side-steps the approximation error a surrogate gradient injects into a transplanted Grad-CAM-style method [37], nor ground-truth labels.

The working hypothesis follows from the first two properties:

> **Quantizing the membrane potential degrades the temporal structure of the explanation before quantizing the weights degrades its spatial structure**,

a hypothesis that originates from the principle that the membrane potential is the variable that integrates evidence over time, while a weight error is fixed for a whole forward pass. The hypothesis is falsifiable, and testing it precisely requires quantizing the two targets separately: the membrane arm should displace the temporal centre of mass of the explanation at a *higher* bit-width than the one at which the weight arm lowers its spatial correlation.

1.3 Contributions

To address the research question, it is necessary to employ a measurement tool. However, since the metrics used throughout the work consistently yield values between zero and one–regardless of the input data–establishing a reference baseline is essential. Indeed, before interpreting the final analysis results, one must determine the values these metrics

produce in the absence of variations. The thesis is therefore structured into two main sections: the first dedicated to the design and calibration of the tool, and the second to its operational application to the quantization grid.

The core contributions are therefore:

1. **A hardware-realizable full-precision baseline.** The input resolution of 112 px was chosen based on network health, measured as dead and saturated neuron fractions, and on the resolution of the explanation maps, not on accuracy: at 64 px the last feature map shrinks to 4×4, too coarse for spatial metrics. The membrane decay was also searched only among values realizable with a shift, alone or followed by a subtraction, and the search returned a value of $\beta = 0.75$. Over three seeds the reference reaches 0.9833 ± 0.0014 validation accuracy.

2. **A corrected implementation of the Spike Activation Map.** The decay parameter γ is fixed on the temporal axis it actually governs. The neuron contribution score now sums over strictly earlier time steps ($t' < t$), which removes a constant pedestal that had inflated map correlations.

3. **A calibration of the explanation-divergence metrics** at four levels: numerical noise, continued training with no bits removed, a thirtyfold change in γ, and independently initialized networks of equal accuracy. Spatial metrics are validated only between a model and its own parent, so every quantized model is compared with the full-precision counterpart model of its own seed.

4. **A bit-width characterization of explanation fidelity.** Weight and membrane quantization are applied separately and then together, at 8, 4 and 2 bits and three seeds, alongside a full-precision fine-tuning control.

5. **An analytical energy and memory account** of every operating point, with synaptic operations computed from per-layer firing rates, in order that accuracy, explanation fidelity, and energy can be read on a common bit-width axis.

The central result is a configuration that the accuracy criterion alone cannot discriminate from its neighbours. Quantizing the membrane potential to 2 bits leaves accuracy at parity, 0.982, yet the region its explanations highlight overlaps the full-precision one by 0.44, well below the numerical floor of 0.70; a 4-bit weight model of the same accuracy keeps an overlap of 0.74. The change is real but bounded, as for the classes on which SAM is faithful, the deletion metric shows that the new explanations still point at evidence the model uses, but not at the evidence the original model pointed and looked at.

The hypothesis stated above is ultimately not supported. Read against the floor of each class, the two effects it orders appear at the same bit-width: at 4 bits the membrane arm moves the temporal centre of mass above its floor, and the weight arm lowers the spatial correlation below its floor in all four classes, while moving the centre of mass by a similar amount in the opposite direction. At 2 bits the membrane modifies *where* the explanation points more than *when*. The mechanism behind the hypothesis is supported, as the same quantization makes the inter-spike intervals of the first layer exactly regular: the coefficient of variation measured within an image is zero for every neuron of every image, in all six models with a 2-bit membrane, over the 20 to 53 % of neuron-image pairs that produce the two intervals a coefficient needs. The three full-precision parents, measured the same way, return 0.153 to 0.188. The temporal metric of the explanation does not register this change.

The configuration with the changed explanation is also dominated on energy and memory, so in this network there is no trade-off to make. A joint 4-bit model keeps accuracy at parity and its explanation at the numerical floor, with one eighth of the memory and about 1/60 of the estimated energy if the input image is quantized to the same width as the weights; if the first layer stays in floating point, the factor is 1/2.8.

1.4 Application domain

The classification task is the identification of transient noise — known as *glitches* — in the data of the LIGO gravitational-wave detectors, using the Gravity Spy release [44, 16, 15].

The domain was chosen for three reasons. Explanation fidelity particularly has an *operational* consequence: glitch classification guides detector characterization, where a commissioner can act physically on the instrument: a compressed model reaching the same label from different evidence would preserve the label but would be corrupting the diagnostic cue.

The classes have physically interpretable morphology, so a domain expert can judge whether an attention map falls where the glitch actually is, which is not necessarily the case of a generic image benchmark. Moreover, the data is public and large, with every trigger labelled by the Gravity Spy classifier. The labels are machine labels (not a verified ground truth), and in Section 2.4 it is reported how far the scores released with them support them.

1.5 Structure

Chapter 2 reviews spiking neural networks, their training by surrogate gradients, quantization for spiking architectures, explainability methods applicable to them, and the gravitational-wave classification task. Chapter 3 describes the data pipeline, the model, the selection of the operating configuration, the Spike Activation Map and the protocol used to calibrate the divergence metrics. Chapter 4 describes the quantization study: the quantizers, the grid, the fine-tuning protocol, the divergence metrics and the energy model. Chapter 5 reports the full-precision reference, the physical validation of the network dynamics, the explainability analyses, the calibration, and the results of the quantization grid. Chapter 6 discusses the findings and their limits, and outlines future work.

Chapter 2
Background and Related Work

2.1 Spiking neural networks

Spiking neural networks (SNNs) are often described as the third generation of neural networks [27]. They take their inspiration from biological neurons, which communicate through brief, all-or-nothing electrical impulses called *spikes*. In a conventional ANN, each neuron outputs a continuous value computed once per input, while in an SNN, information is carried by sequences of binary events distributed over time: at each of T discrete time steps, a neuron either emits a spike or stays silent. The network therefore processes an input as a short temporal sequence. This holds even when the input is static, as with a spectrogram, our specific case, which is presented repeatedly or encoded as spikes over the T steps.

This representation is exactly what makes SNNs attractive for low-power computing. Because activations are binary, the multiply–accumulate operations that dominate the cost of an ANN reduce to simple accumulations, performed only when a spike actually is fired. Computation is therefore sparse and event-driven, with a silent neuron costing almost nothing. Neuromorphic processors such as Intel's Loihi [10] are built to exploit this sparsity directly, and the energy advantage of an SNN is set to grow as its firing activity decreases.

The most widely used neuron model in deep SNNs is the leaky integrate-and-fire (LIF) neuron [14]. In this configuration, each neuron maintains an internal state, its membrane potential U, which integrates incoming input, decays ("leaks") towards rest over time, and triggers a spike when it crosses a threshold θ.

As shown in Figure 2.1, the LIF model sits between biologically detailed neuron models and the artificial neurons of conventional deep learning (ANNs).

In the discrete-time formulation used by snnTorch [12], the dynamics of a neuron at time step t are the following

$$U[t] = \beta\, U[t-1] + WX[t] - S[t-1]\,\theta, \tag{2.1}$$

$$S[t] = \begin{cases} 1 & \text{if } U[t] > \theta, \\ 0 & \text{otherwise,} \end{cases} \tag{2.2}$$

where $WX[t]$ is the weighted input from the previous layer, $S[t]$ is the binary output spike, and the subtraction term resets the potential by θ after each spike. The decay factor $\beta \in [0, 1)$ sets how much of its past state the neuron retains: values close to 1 give a long memory, so the neuron integrates evidence over many time steps; smaller values make it respond mainly to recent input.

The property of the membrane potential to give the LIF neuron a memory is what separates it from a stateless activation function: the output at time t depends on the entire history of inputs up to t. The same mechanism, nonetheless, also creates the main difficulty in training. The spike function in Eq. (2.2) is a step function, whose derivative is zero

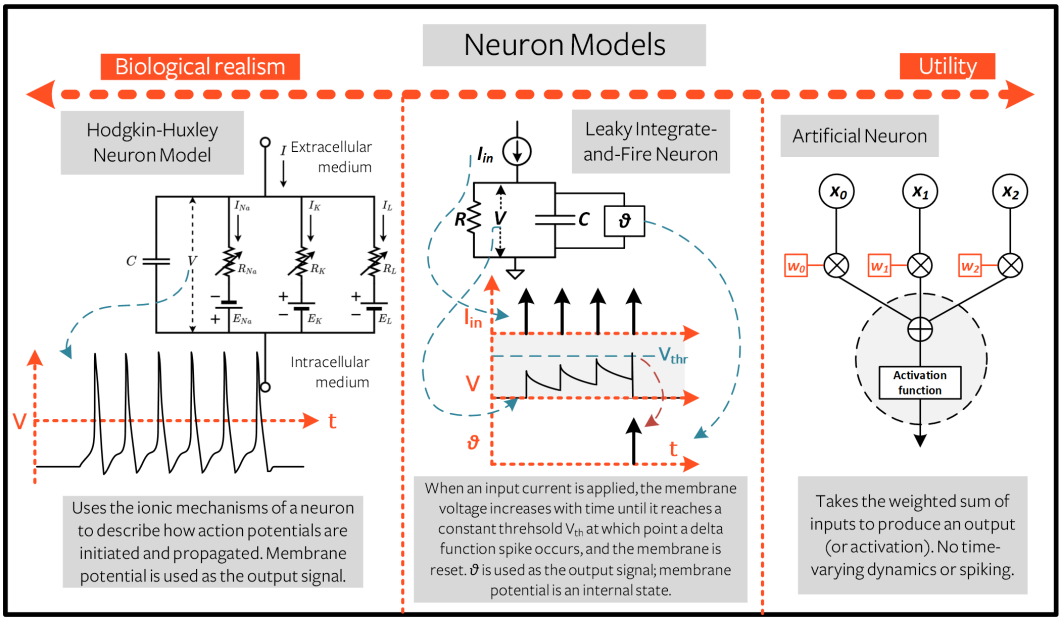


Figure 2.1: Neuron models ordered from biological realism to computational utility. Left: the Hodgkin–Huxley model describes the generation of action potentials through voltage-dependent ionic conductances, at a high computational cost. Centre: the leaky integrate-and-fire (LIF) neuron reduces the membrane to an RC circuit: the capacitor C integrates the input current I_{in}, while the resistor R makes the stored charge leak away over time, and a spike is emitted when the membrane potential reaches the threshold (denoted ϑ in the figure and θ in the text), after which the potential is reset. Unlike the Hodgkin–Huxley model, the membrane potential of a LIF neuron is an internal state and the spike is the output. Right: the artificial neuron of conventional ANNs, which computes a weighted sum of its inputs followed by an activation function, with no temporal dynamics and no spikes. Reproduced from the snnTorch documentation [11] (MIT License).

almost everywhere and undefined at the threshold, so standard backpropagation cannot pass through it. The usual solution is the *surrogate gradient* approach [30]: the step function is kept in the forward pass, but in the backward pass its derivative is replaced with that of a smooth approximation, such as an arctangent. Combined with backpropagation through time over the T steps, this allows deep SNNs to be trained with the same gradient-based tools as conventional networks.

These properties change the quantization problem in three main direct ways. First, SNN activations are already binary, so the precision that quantization removes lies in the weights and in the membrane potential, a state variable with no direct equivalent in an ANN. Second, the threshold makes the network highly sensitive to small numerical changes. Indeed, a perturbation that moves U slightly above or below θ turns a spike into silence, or the reverse. Because the membrane carries its state forward, such an error does not stay local, but can propagate through later time steps and layers. Third, explanations of an SNN can be temporal as well as spatial. The Spike Activation Map (SAM) [22] builds attribution maps directly from the timing of spikes, without computing gradients, and therefore shows not only *where* the network finds evidence in the input but also *when* during the T time steps it does so. Whether quantization preserves this temporal reasoning, and not only accuracy, is the question this thesis sets out to answer.

2.1.1 Training by surrogate gradients

Unrolled over its T time steps, a network of LIF neurons is effectively a recurrent network: the membrane potential is a hidden state that each step reads and updates (Equation 2.1). Training it by gradient descent requires solving two credit-assignment problems at once [30]: a *spatial* one, across layers, as in any multi-layer network, and a *temporal* one, since a weight influences the loss through every later state of the neurons it feeds. Backpropagation through time (BPTT) solves both by applying the chain rule to the unrolled graph. For a weight W feeding a neuron with potential $U[t]$ and output $S[t]$,

$$\frac{\partial \mathcal{L}}{\partial W} = \sum_t \frac{\partial \mathcal{L}}{\partial S[t]} \frac{\partial S[t]}{\partial U[t]} \frac{\partial U[t]}{\partial W}, \qquad \frac{\partial S[t]}{\partial U[t]} = \Theta'(U[t] - \theta), \tag{2.3}$$

where Θ is the Heaviside step of Equation 2.2 and $\partial U[t]/\partial W$ is the total derivative through all earlier states. The price is memory: every state of every time step must be kept until the backward pass, a cost that grows as $O(NT)$ for a layer of N neurons.

The obstacle. Every path through Equation 2.3 contains the factor Θ', which is zero everywhere except at the threshold, where it is undefined. The gradient is therefore either zero or meaningless: a small change of a weight almost never flips a spike, so the loss is piecewise constant in the weights, with flat plateaus on which gradient descent cannot move. This is the same problem met by networks with binary activations, and the solutions developed for spiking networks are inspired by those [30].

The surrogate. A surrogate gradient leaves the model unchanged and alters only its derivative. The forward pass keeps the exact binary spike, while in the backward pass every occurrence of Θ' is replaced by the derivative σ' of a smooth function peaked at the threshold,

$$\frac{\partial S[t]}{\partial U[t]} \longrightarrow \sigma'(U[t] - \theta). \tag{2.4}$$

Neurons close to threshold, meaning those for which a small change could flip a spike, receive the largest gradient, and gradient flows through silent neurons as well. The resulting update is the gradient of a *virtual* surrogate loss, never computed explicitly, which follows the true one while remaining continuous and finite [30]. This distinguishes the approach from *smoothed* spiking networks, which involve replacing the spike itself with a soft or probabilistic function and so change the model that is deployed.

The choice of surrogate. Piecewise-linear, fast-sigmoid and exponential surrogates have all been used successfully, all three nonlinear and monotonically increasing towards the threshold. The diversity of working choices suggests that the method does not depend critically on the exact shape [30]. This work uses the arctangent surrogate [13] as implemented in snnTorch [12],

$$\sigma'(x) = \frac{\alpha/2}{1 + \left(\frac{\pi}{2}\,\alpha\, x\right)^2}, \qquad \alpha = 2, \tag{2.5}$$

which equals 1 at the threshold and falls off as the inverse square of the distance from it, so that neurons far from threshold still receive a small, non-zero gradient. The reset term $-S[t-1]\,\theta$ of Equation 2.1 is treated as a constant in the backward pass: the recurrence then contributes $\partial U[t]/\partial U[t-1] = \beta$ alone, without a second surrogate factor through the reset, a simplification that has been found empirically to improve training [30]. With this training configuration, spiking networks can manage to reach accuracies comparable to conventional architectures on image benchmarks [12].

The same device reappears in Section 2.2: the straight-through estimator used to train through a rounding operation [7] replaces the zero derivative of a step function with that of a smooth one, exactly as the surrogate gradient does for the spike.

A consequence for explainability. The substitution motivates the choice of explanation method in Section 2.3: any gradient-based attribution computed on such a network inherits the approximation error of the surrogate, because the gradients it uses are not the gradients of the function the network actually computes.

2.1.2 Input encoding

A spiking network processes its input over T discrete time steps, while an image is a single static frame, so the frame must be turned into a sequence. In this context, two strategies dominate. In *rate coding* each normalized pixel intensity is used as the probability of emitting a spike at every step, and the value is carried by the spike count; in *direct* (or *constant*) encoding, the analog frame itself is presented, identical, at every time step.

Rate coding pays its simplicity in time steps: a pixel is represented by counting spikes, so with T steps it can take only $T+1$ distinct values, and the sampling makes the representation stochastic: the discretization error shrinks only as T grows, which is why rate-coded and converted networks typically require between 100 and 2 000 time steps to approach the accuracy of their non-spiking counterparts [34].

Direct encoding, as introduced by DIET-SNN [34], removes that error at the source. "The pixel intensities of an image are applied directly to the input layer of the SNN at each timestep", and the first convolutional layer, followed by LIF neurons, becomes at once a feature extractor and a learned spike generator: it integrates the weighted pixel values and fires when its membrane potential crosses threshold. Since the input carries its full precision at every step, the number of steps is no longer bounding the precision of the input. DIET-SNN combines this encoding with a trained membrane leak and firing threshold per layer, which increase activation sparsity, and reaches 69 % top-1 accuracy on ImageNet with 5 time steps and about 12× less compute energy than the equivalent conventional network [34]. This work adopts the encoding but not the trained neuron parameters: β and the threshold are shared by all layers, and β is chosen by the constrained search of Section 3.4.2.

Direct encoding has three remarkable consequences that the rest of this thesis relies on.

Temporal structure is generated inside the network. Since the stimulus does not vary, **all temporal structure in the network originates from the internal LIF dynamics** — leak, integration and reset — and not from variation in the input. The time axis of a spike

train, and therefore of a Spike Activation Map, is the time the network takes to integrate evidence, which is not the time of the glitch: in an Omega-scan spectrogram, physical time runs along the horizontal axis of the image and is, for the network, a spatial dimension. Every temporal analysis in this thesis is to be read in that light.

The first layer is the only one that multiplies. Every other layer receives binary spikes and only accumulates, but the first convolution receives analog pixel values and performs full multiply-accumulates. Because its input is identical at every step, so is its output current, which can be computed once rather than T times. Section 4.8 shows how this choice decides whether the spiking network is cheaper than its conventional equivalent.

The input is deterministic. Without sampling, two forward passes on the same image produce identical spike trains, which is what allows two models to be compared spike for spike on the same input, as the calibration of Section 5.6 requires.

2.2 Quantization

Quantization represents the parameters and intermediate values of a network with fewer bits than the 32 of floating point. Storing a value in b bits divides its memory by $32/b$, and integer operations on narrow operands cost far less energy than floating-point ones (Section 2.2.2), which is why quantization is the standard practice adopted between a trained model and its deployment on constrained hardware [20, 29].

The uniform quantizer. A uniform quantizer maps a real value x to one of at most 2^b equally spaced levels: it divides by a step Δ, rounds to the nearest integer, clips to the representable range, and multiplies back,

$$\hat{x} = \Delta \cdot \mathrm{clip}\Big(\mathrm{round}\Big(\frac{x}{\Delta}\Big),\, q_{\min},\, q_{\max}\Big) \tag{2.6}$$

where $q_{\min}$ and $q_{\max}$ are the smallest and largest integer levels representable with b bits: $-(2^{b-1}-1)$ and $2^{b-1}-1$ for a symmetric signed grid, which keeps zero exactly representable at the cost of one unused code, 2^b-1 levels in all; 0 and 2^b-1 for an unsigned one, which uses all 2^b. The largest representable value is therefore $\Delta\, q_{\max}$.

For a symmetric grid over $[-w_{\max}, +w_{\max}]$ the largest level must reach $w_{\max}$, so the step is

$$\Delta = \frac{w_{\max}}{2^{b-1}-1} = \frac{2w_{\max}}{2^b-2}, \tag{2.7}$$

the step the weight quantizer of Chapter 4 computes for each output channel. Section 5.6.3 expresses a perturbation in a different unit: the 8-bit step of a single global grid, set by the largest weight of the whole network and computed there as $2w_{\max}/(2^8-1)$. At equal $w_{\max}$ the two differ by 0.4 %, one unit in the denominator. The operative difference is the scope rather than that arithmetic, since a grid set by the largest weight of the network is an upper bound on the per-channel steps actually applied (Table 4.2).

Since the quantizer works per channel, and most channels have a smaller maximum than the network as a whole, that unit is coarser than the steps the quantizer actually applies, and the floor it defines is conservative.

Two main errors follow from Equation 2.6, with the choice of range trades one against the other: values inside the range are moved by at most $\Delta/2$ (*rounding* error), values outside it are cut to the nearest bound (*clipping* error). For values spread over many steps the rounding error is close to uniform on $[-\Delta/2, \Delta/2]$, with a root-mean-square of $\Delta/\sqrt{12}$; a Gaussian perturbation whose standard deviation is one step, the unit of Section 5.6.3, is therefore about 3.5 times larger than the rounding it stands for. A wide range avoids clipping but makes Δ coarse; a narrow one refines Δ at the cost of saturating the tails.

When the quantity is non-negative, or its negative values carry no information for the task, an *unsigned* grid over $[0, x_{max}]$, with step $x_{max}/(2^b - 1)$, spends no levels on values the network does not use. The membrane potential of this network is of the second kind. Negative potentials are frequent — in the first layer the median is -0.08 and the first percentile -6.6 — yet clipping them to zero moves the validation accuracy by one image in 1 435 (Section 5.7).

More specifically, the step can be shared by a whole tensor (*per-tensor*) or computed separately for each output channel (*per-channel*). A per-tensor step is set by the largest value in the tensor, so channels whose weights are systematically smaller use only a fraction of the available levels; the per-channel scheme avoids this at the price of one scale factor per channel [24, 29].

Given a quantizer, two families of method decide *when* it is applied: after training, or during it.

Post-training quantization (PTQ). PTQ converts an already trained full-precision model: the weights are rounded to their grid, and the ranges of the activations, which depend on the data, are estimated on a small unlabelled calibration set [29]. It is a relatively fast method and needs neither labelled data nor a training pipeline, which makes it the "push-button" option and the one most used for rapid deployment. Its limit is precision: at 8 bits the rounding error is small compared with the spread of the weights, and per-channel PTQ keeps classification accuracy within about 2 % of floating point on standard convolutional networks. At 4 bits the step is sixteen times coarser, and accuracy typically degrades unless more elaborate correction methods are used [24, 29]. Rabbi et al. [33] use static weight-only PTQ at INT8 and INT4, the setting in which they observe explanations degrading before accuracy.

Quantization-aware training (QAT). QAT includes the quantizer in training, so that the network can adapt its weights to the grid. Specifically, in the forward pass every quantized tensor is passed through Equation 2.6 — *simulated* or *fake* quantization, since the values are rounded but still stored and processed in floating point — so that the loss is computed on the network as it will be deployed [20]. The difficulty lies in the backward pass: rounding is a step function, whose derivative is zero almost everywhere, and no gradient would reach the weights. The *straight-through estimator* solves it by treating rounding as the identity in the backward pass [7]: the forward pass sees the quantized value, the gradient flows as if it had not been quantized. This is the same device, in a different guise, as the surrogate gradient of Section 2.1.1, which also keeps a step function in the forward pass and replaces its derivative in the backward one. QAT needs labelled data and additional training, usually as a fine-tuning of the full-precision model, and in exchange reaches lower bit-widths: it narrows the 8-bit gap to about 1 % and makes 4-bit weights usable, with losses between 2 and 10 % depending on the size of the network [24, 29].

The two families therefore answer different needs: PTQ when the model must be compressed cheaply and 8 bits suffice, QAT when the target precision is lower than PTQ can reach. In a spiking network both apply not only to the weights but also to the membrane potential, as the next section describes.

2.2.1 Quantization for spiking networks

Spiking architectures add constraints of their own, as could already be inferred. The membrane decay β multiplies a stored state at every time step, so on integer hardware it is cheapest when realizable as a bit shift: $\beta = 2^{-k}$ is a right shift, and $\beta = 1 - 2^{-k}$ is a shift plus a subtraction. MINT [43] quantizes both targets uniformly and shares a single scale factor between weights and membrane potentials. Since the weighted input and the stored state are then expressed in the same integer units, they can be added directly, and the multipliers that conventional uniform quantization needs to rescale one into the other disappear from the inference loop. At 2 bits, MINT reports 90.6 % accuracy on CIFAR-10

with a VGG-16 and a memory reduction of 93.8 % with respect to full precision. HardF-SNN [26] shows the cost of that sharing: weights have a narrow, centred distribution, whereas membrane potentials have a much broader one with a long tail, so a scale fitted to the weights compresses the dynamic range of the membrane and collapses many distinct potentials onto the same level. It therefore scales the membrane in proportion to the weights rather than identically, replaces batch normalization with an integer, shift-based version, and verifies the result on an FPGA accelerator, noting that most quantized spiking networks are still evaluated with simulated quantization, in which part of the computation remains in floating point.

Smith et al. [38] argue independently that accuracy is an insufficient criterion for quantized spiking networks. They in fact propose an Earth Mover's Distance between per-neuron firing-rate distributions as a diagnostic of behavioural divergence. They separate weight from membrane quantization as this thesis does, and report that membrane-potential quantization requires at least four bits to preserve both accuracy and dynamics while learned weight quantization remains viable at two. They ground the asymmetry mechanistically: a weight quantization error is fixed for an entire forward pass, whereas a membrane error propagates through the recurrent update at every time step.

This thesis reaches the opposite result at 2 bits: the membrane arm keeps accuracy at parity (Section 5.9). A plausible reason is where the quantizer acts. Here it rounds the state carried into the next step, after the spike decision (Section 4.4). At every step the threshold is compared with β times the quantized previous state plus a full-precision input current, so the comparison itself is never made on a 2-bit value.

2.2.2 Energy and memory accounting

Since no model is deployed on hardware in this work, energy is estimated analytically, as is standard in the spiking literature: the operations an inference requires are counted and multiplied by a per-operation cost at a reference process node, conventionally the 45 nm figures of Horowitz [17]. The result is a like-for-like comparison between configurations of the same network.

Counting operations. In a conventional layer every connection performs one multiply-accumulate (MAC) per inference. In a spiking layer the input is binary, so a connection only *adds* its weight, and only when the presynaptic neuron fires. The number of these synaptic operations (SOP) is therefore the dense MAC count scaled by the firing rate f of the input layer and by the number of time steps,

$$\mathrm{SOP} = f \cdot T \cdot \mathrm{MAC}. \tag{2.8}$$

In 32-bit floating point a MAC costs 4.6 pJ (a 3.7 pJ multiplication plus a 0.9 pJ addition) and an accumulate 0.9 pJ. That factor of five is the physical reason a spiking network can be cheaper, and it also sets the limit: a spiking layer saves energy only while $f \cdot T < 4.6/0.9 \approx 5$, so sparsity and a short time window are both necessary. The first layer is an exception, since under direct encoding it receives analog values and performs full MACs (Section 2.1.2).

Quantized operations. The same table supplies the integer figures used for quantized operating points: an 8-bit multiplication costs 0.2 pJ and an 8-bit addition 0.03 pJ, thirty times less than its floating-point counterpart, so that an 8-bit MAC costs 0.23 pJ under the same convention that gives 4.6 pJ in floating point. Narrower integer operations are not tabulated, so any figure below eight bits is an extrapolation, whose rule is stated in Chapter 4.

What the count leaves out. The estimate covers arithmetic only. It ignores the update of each neuron (leak, threshold comparison, reset), control logic, and above all data movement, which on real hardware is often the larger cost: in the same 45 nm table, reading

64 bits from on-chip SRAM costs between 10 and 100 pJ depending on its size, and from off-chip DRAM between 1.3 and 2.6 nJ [17], one to three orders of magnitude more than an arithmetic operation.

Why memory is accounted separately. For this reason the memory footprint is reported alongside the energy estimate: the energy of an access depends on where the data lives, which is a property of the hardware, whereas the number of bits to be stored is a property of the model. The footprint is computed as the bits needed for the weights plus the bits needed for the membrane state of every neuron. The second term is specific to spiking networks: a conventional network can discard the activations of a layer once the next one has been computed, but a spiking network must keep the potential of every neuron from one time step to the next for the whole inference. That state scales with the number of neurons rather than the number of parameters, and it can be quantized independently of the weights (Section 2.2).

2.3 Explainability for spiking networks

Attribution methods developed for conventional networks transfer badly to spiking ones, for two different causes. Gradient-based methods such as Grad-CAM [37] weight feature maps by the gradient of the class score, and in a spiking network that gradient exists only as a surrogate (Section 2.1.1). The approximation is applied again at every layer the gradient crosses, so the error accumulates with depth: Kim and Panda [22] observe that Grad-CAM transplanted to spiking networks produces smoothed, poorly discriminative maps, worst at the shallow layers. Perturbation methods such as LIME [35] avoid gradients altogether and are architecture-agnostic, but they need hundreds to thousands of forward passes per explanation. Both families, moreover, return a single map per input, and so discard the temporal axis along which a spiking network computes.

The Spike Activation Map. The Spike Activation Map (SAM) of Kim and Panda [22] was designed for spiking networks from the start. It rests on a biological observation: spikes that follow each other at short inter-spike intervals are more likely to drive the membrane of the downstream neuron above threshold, and therefore carry more information than isolated ones. SAM numerically quantifies this into a score computed in three steps. A *temporal spike contribution score* weights a past spike at t' by its recency with respect to the current step t,

$$T(t, t') = \exp\left(-\gamma\,|t - t'|\right), \tag{2.9}$$

where γ sets how fast the contribution of a spike fades. A *neuronal contribution score* sums these weights over the spikes the neuron emitted before t,

$$N^{k}_{ij,t} = \sum_{t' \in P^{k}_{ij},\, t' < t} T(t, t'), \tag{2.10}$$

where P^{k}_{ij} is the set of firing times of the neuron in channel k at position (i, j); it is high for a neuron that has fired often and recently, and low for one that fired rarely or long ago. Finally, the map at time t keeps the score only of the neurons that are firing at t and sums over the channels of the layer,

$$M_{ij,t} = \sum_{k} N^{k}_{ij,t}\, S^{k}_{ij,t}, \tag{2.11}$$

yielding one spatial heat map per time step and per layer.

Properties. Several properties of SAM matter for this thesis. It uses solely the spike trains of the forward pass, so the explanation describes the function the network actually computes, and its cost is marginal — a store of recent spike times and a lookup table for the kernel, which Kim and Panda note could be implemented in an inference accelerator. It needs no class label: SAM shows where the network concentrates its activity for a given input, not the evidence for a particular class, so its faithfulness must be tested against the class the model predicts. And it has a temporal axis: an explanation is a sequence of T maps, which is what makes it possible to ask whether quantization changes *when* a network decides as well as *where* it looks. Kim and Panda validate the method on Tiny-ImageNet, finding SAM more discriminative than Grad-CAM adapted to spiking networks.

Feature-attribution methods for spiking networks have also been surveyed more broadly [31], but SAM remains the method that reads the spike trains directly, and it is the explainer used throughout this work. Its one free parameter, γ, is fixed in Section 5.5.1.

2.3.1 Evaluating an explanation

A stable explanation can still be wrong, so stability requires a separate test of faithfulness. The *deletion metric* of Petsiuk et al. [32] removes pixels in decreasing order of attributed importance, and records how quickly the model's confidence falls, summarized as the area under that curve. The faster the fall, the smaller the area. The underlying hypothesis is that a faithful map degrades the prediction faster than a map that carries no information. Petsiuk et al. read the area on its own; this thesis compares it with the removal of pixels in random order on the same image (Section 5.5.2), so that a map counts as faithful only by the margin it gains over chance.

The metrics used to quantitatively compare two explanations with each other are borrowed from the saliency-model literature, where they were specifically developed to compare a predicted map against human eye fixations [8]: Pearson correlation, structural similarity [42], and the intersection-over-union of the highest-scoring regions. Bylinskii et al. observe that rankings produced by different metrics frequently disagree, which is the main reason for reporting more than one. This thesis reports the correlation and the overlap. Structural similarity, which Rabbi et al. [33] use alongside them, is left out as redundant with the two.

A separate line of work asks whether attribution methods are testing anything at all. Adebayo et al. [4] randomize model weights progressively and verify that explanations change in response, as a map that survives the destruction of the model it sets to explain is not explaining it. The perturbation experiment of Section 5.6.3 is a graded form of that test, run in the opposite direction.

2.4 Gravitational-wave glitch classification

The sensitivity of the LIGO interferometers is limited by non-Gaussian transient noise of instrumental and environmental origin. As introduced in Chapter 1, these glitches are short, can mimic astrophysical signals, and can be grouped into morphological families that correlate with distinct physical causes.

The Gravity Spy project [44, 6] combines machine classification with citizen science to label them. The release used here [15] covers observing runs O1 through O3b and provides, for each Omicron trigger, the trigger metadata, the per-class scores of a convolutional classifier, the label that classifier assigns, and the addresses of the Omega-scan spectrograms, which are downloaded separately from the Zooniverse archive. The labels used in this work are therefore machine labels, not volunteer consensus, and no threshold on the classifier's confidence was applied when the data were selected.

The scores released with the labels do not always support them. Of the 9 568 triggers selected for the four classes, the score of the assigned class is below 0.5 for 24 % and below

0.01 for 615, and for 4.4 % the label is not the class with the highest of the four scores. The disagreement is concentrated in two classes: the score falls below 0.5 for 53 % of the Violin_Mode triggers and 37 % of the Extremely_Loud ones, against 4 % of Blip and 3 % of Scattered_Light. An accuracy measured against these labels is an agreement with the Gravity Spy classifier, not with a verified truth.

The classification task is therefore image classification over time–frequency representations. Its interest for this thesis is not its difficulty — at 98 % agreement with the reference classifier it is not difficult — but that the explanations have a consumer. Glitch classification is directly connected with detector characterization, in which the morphology of a glitch family is traced back to an instrumental cause and the instrument is modified accordingly. An explanation that points at the wrong part of the spectrogram sends a commissioner to the wrong causes.

2.5 The gap this thesis occupies

There are two close works useful to frame this thesis. Rabbi et al. [33] measure explanation fidelity under quantization, but for conventional networks, implementing post-training weight-only quantization, and relying purely on spatial explanations. Smith et al. [38] address spiking networks and separate the two quantization targets, but measure divergence in *population firing statistics* aggregated over a dataset, and do not take into account explanations.

The unoccupied ground is therefore a per-sample, spatially and temporally resolved explanation fidelity for a spiking classifier with the two quantization targets separated.

Chapter 3
Methodology

This chapter describes the data pipeline, the model, the procedure by which the operating configuration was selected, and the implementation of the explainability metrics. Every measurement reported here is on the validation partition.

3.1 Data

3.1.1 Acquisition

The Gravity Spy machine-learning release [15] is distributed as eight CSV files, one per detector (H1, L1) and observing run (O1, O2, O3a, O3b). Each record corresponds to an Omicron trigger with signal-to-noise ratio above 7.5 and peak frequency between 10 Hz and 2048 Hz, and carries the trigger metadata, the per-class confidence of the Gravity Spy classifier, the assigned label, and URLs to Omega-scan spectrograms.

The labels are the classifications produced by that convolutional model, not a physical ground truth and neither the volunteer-combined label of the separate citizen-science release. For this reason, accuracy reported in this thesis is in agreement with that reference classifier.

3.1.2 Class selection

The original dataset featured twenty-four classes, of which two are not glitch morphologies: `No_Glitch`, the absence of a morphological trigger, and `None_of_the_Above`, the classifier's residual category. Only four classes were retained — **Blip**, **Extremely_Loud**, **Scattered_Light**, **Violin_Mode** — on three criteria established before any separability analysis, the successive multivariate checks are therefore a verification rather than the selection rule.

Abundance. These are all sufficiently populated classes, with 29 573, 24 002, 99 665 and 2 392 members respectively. The smallest sets the balanced cardinality at 2 392 per class, sufficient to train without augmentation (Figure 3.1).

Morphological diversity. The four retained classes span physically distinct families of instrumental noise [44, 16, 6]: short broadband transients of teardrop morphology and ∼0.04 s duration (Blip) [9]; high-amplitude events that saturate the spectrogram (Extremely_Loud); long-duration arches of 2–2.5 s correlated with ground motion, caused in O3 by relative motion between the optical suspension and reaction-mass chains (Scattered_Light) [40]; and narrowband lines at ∼500 Hz and harmonics, corresponding to resonances of the glass fibres suspending the mirrors (Violin_Mode). The choice of spanning the morphological range serves two purposes: it preserves class separability, which keeps the classification problem itself manageable, since it is not the main scope of

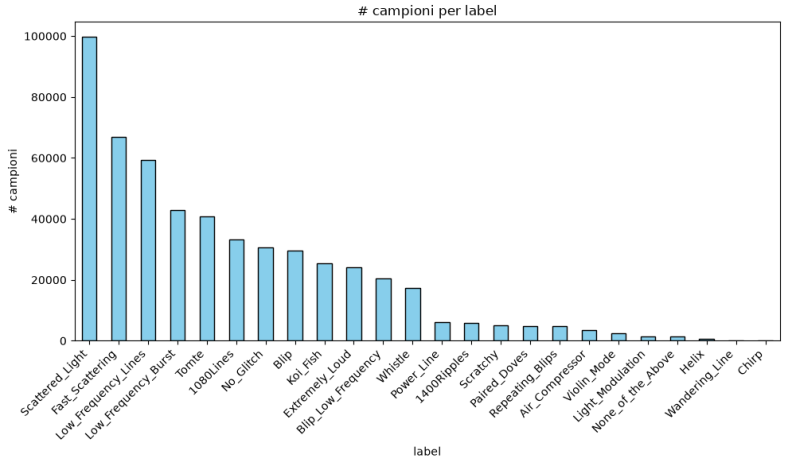


Figure 3.1: Class frequencies in the Gravity Spy machine-learning release, all observing runs and both detectors combined. The distribution is strongly imbalanced.

the thesis, and makes the explanation analysis informative, since each class should elicit a recognizably different attention pattern. An example of each is shown in Figure 3.2.

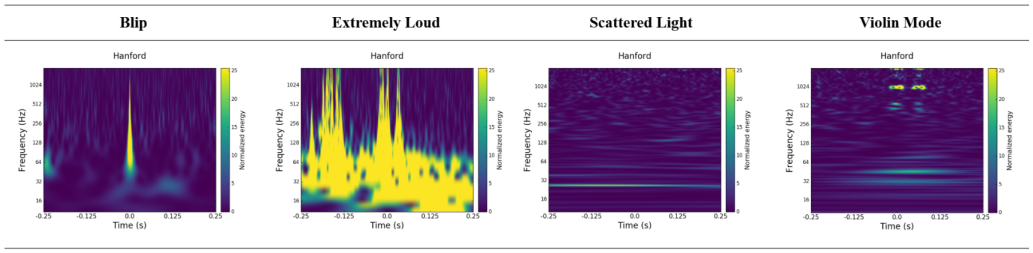


Figure 3.2: One Omega-scan spectrogram per retained class (Hanford detector, 0.5 s window). From left: a short teardrop-shaped transient, a saturated broadband event, a long low-frequency band, and narrow lines near 500 Hz and 1 kHz.

Practical constraint. Four classes keep training and the downstream quantization grid feasible on a single 6 GB consumer GPU.

Verification in the tabular domain supported the choice, as it shows that each class is isolated by a *different* descriptor, which is the relevant property: Extremely_Loud occupies $\log(\mathrm{SNR}+1)$ between 6 and 12 while the others lie between 2 and 5; Blip is separated by duration; Scattered_Light sits below roughly 100 Hz in peak frequency with a bimodal bandwidth distribution; and Violin_Mode is the only class extending to 2 kHz (Figure 3.3). A UMAP projection [28] of the standardized descriptors, computed on a stratified subsample of 500 events per class ($n_{\mathrm{neighbors}} = 15$, seed 42), confirms the separation independently (Figure 3.4).

Violin_Mode is the weakest case, and this nature along with its conformation anticipates a result that recurs throughout the thesis analysis and results: in the $\log(\mathrm{SNR}+1)$ versus $\log(\mathrm{duration}+1)$ plane it overlaps both Blip and Scattered_Light, being separated only in peak frequency. It will indeed be the lowest-recall class at every configuration tested and the most sensitive class in every explainability measurement.

As a side but dutiful note, restricting the problem to four classes keeps it tractable, but means the results are not directly comparable to the full Gravity Spy benchmark.

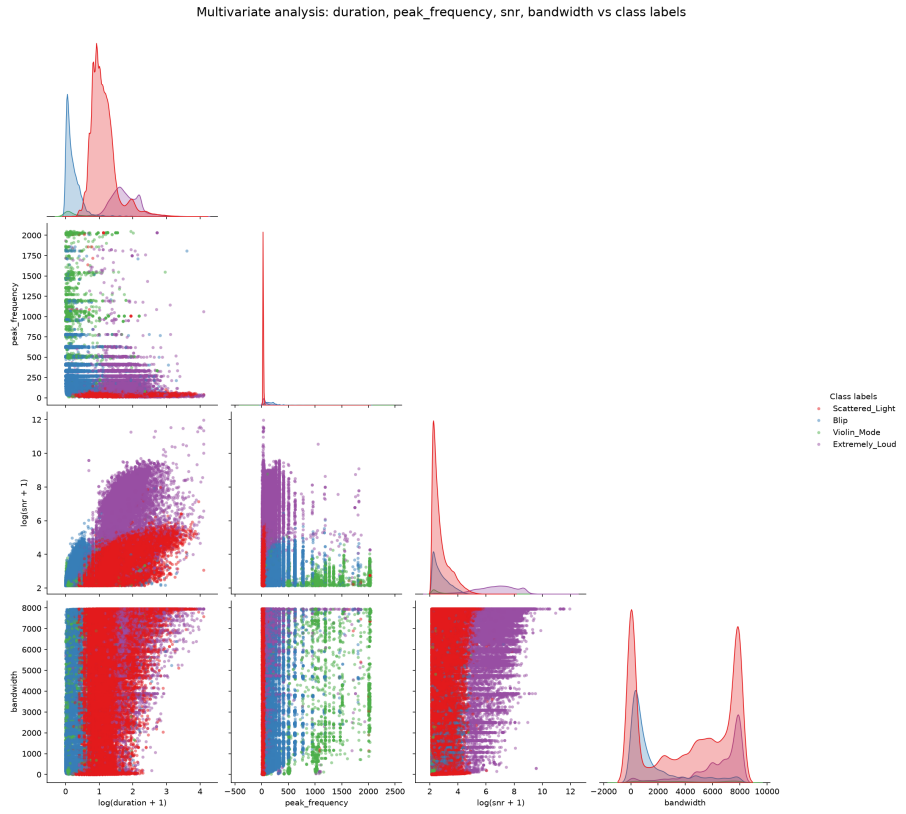


Figure 3.3: Pair plot of the four retained descriptors, coloured by class. Each class is isolated by a different descriptor; Violin_Mode overlaps Blip and Scattered_Light except in peak frequency.

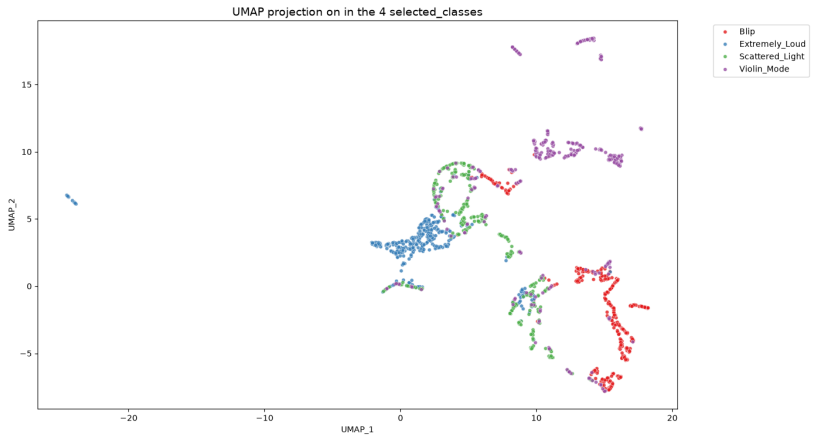


Figure 3.4: Two-dimensional UMAP projection of the standardized descriptors, 500 events per class.

3.1.3 Feature selection

Of the twelve numerical Omicron descriptors, only four were kept and retained indicative. The feature selection was mainly operated based on four distinct criteria, direved from their distributions and from their correlation matrix (Figure 3.5).

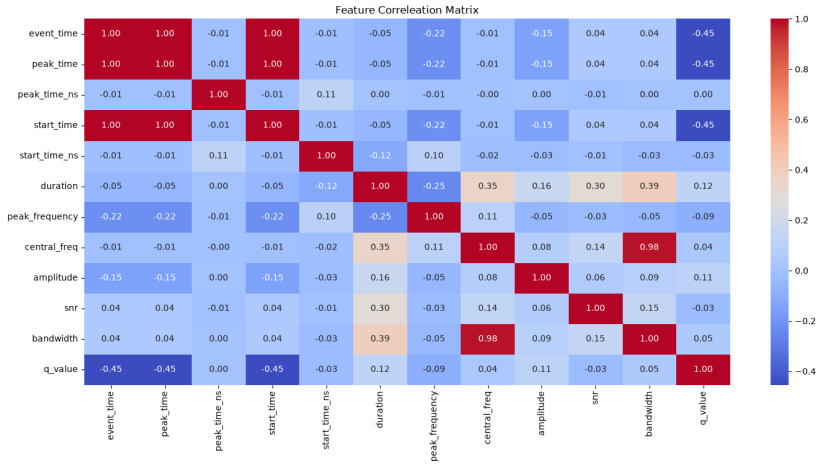


Figure 3.5: Pearson correlation between the twelve numerical Omicron descriptors. The three timestamps are perfectly correlated, and central frequency and bandwidth correlate at 0.98.

Exact redundancy. The three timestamp variables (event time, peak time, and start time) are mutually correlated at $r = 1.00$, being the same GPS time to within sub-second offsets, so at most one of them could be retained. Central frequency and bandwidth correlate at 0.98; bandwidth was retained because peak frequency already encodes spectral position, so bandwidth supplies extent rather than duplicating location.

Temporal provenance. Redundancy leaves one timestamp admissible, but none is retained. A timestamp supplies *when* a glitch occurred rather than *what* it is, and the correlation matrix shows that the shortcut is available: event time correlates at -0.22 with peak frequency, reflecting instrumental drift across observing runs. A model given the time of the trigger could separate the classes by observing period instead of by morphology, and that separation would not transfer to data recorded later.

Degenerate distributions. The nanosecond remainders are uniformly distributed and no correlation of theirs exceeds 0.12 in absolute value, consistent with timing quantization noise, while the quality factor contains a sentinel value of order 10^{20} that compresses the real distribution to the axis origin.

Reference to the noise floor. Amplitude and SNR measure the strength of the same event on two scales. Omicron reports amplitude as a strain amplitude of the trigger; the SNR is computed on whitened data and is therefore that amplitude referred to the noise spectral density of the detector at the frequency and time of the event, $\mathrm{SNR} \sim A/\sqrt{S_n(f,t)}$. The noise curve spans three to four orders of magnitude between 10 and 2048 Hz and differs between H1 and L1 and between observing runs, so across the population the denominator varies more than the numerator. Their linear correlation is 0.06, indeed the two descriptors are not statistically redundant, and neither one is the substitute for the other. Amplitude was dropped because an absolute strain amplitude is not comparable across frequency, detector and epoch, while SNR is comparable and is the quantity on which the release itself is thresholded, at 7.5. Amplitude also correlates at -0.15 with the three timestamps against 0.04 for SNR, carrying the calibration history of the instrument into the descriptor and reopening a similar shortcut as the one discussed before.

The four retained descriptors — peak frequency, duration, bandwidth, SNR — form a complete and non-redundant characterization of a transient in the time–frequency plane:

spectral position, temporal extent, spectral extent and prominence. Their largest pairwise correlation is 0.39, well below the 0.98 that forced the exclusion of central frequency.

Only one of the four image URL columns was kept. The four correspond to successive time windows of the same detection (0.5 s, 1 s, 2 s, 4 s), and only the first was retained. Because the time windows are effectively *different views*, they are not interchangeable, which is why unavailable images were excluded instead of substituted from another column.

Residual class imbalance is removed by random undersampling to the cardinality of the smallest class, giving an exactly balanced set of 9 568 images.

3.1.4 Preprocessing and partition

The downloaded spectrograms are matplotlib formatted figures comprising a frame with axes, colour bar and annotation, therefore a centre crop of 470×550 pixels isolates the spectrogram panel, which is then resized to the working resolution. The order of operation matters, since cropping the original and resizing afterwards keeps the same portion of the spectrogram framed at every resolution. Channel-wise normalization uses the standard ImageNet statistics, mean $(0.485, 0.456, 0.406)$ and standard deviation $(0.229, 0.224, 0.225)$. These are not the statistics of this corpus, and has a consequence that matters later in the analyses: zero in normalized space is the ImageNet channel mean, and that is the value the deletion metric substitutes for a removed pixel (Section 3.7.2). A removed pixel is therefore set to a neutral grey rather than to black, which would itself be a strong input.

The partition is frozen to disk. A dedicated script generates the train/validation/test assignment once, 70 / 15 / 15 under a fixed seed, and writes it as a per-file record of path, class and split, so that every subsequent run reads that file. A file listed but absent from disk is dropped from its partition with a warning. A file present on disk but absent from the assignment is never loaded, the loader iterating the assignment rather than the directory, so no sample can enter a partition without it having been assigned to one.

Freezing the partition allows for the assignment to be a function of the file list rather than of the corpus size, so removing a file removes exactly that file instead of reshuffling every remaining assignment. And the audit trail is externally verifiable: a reader can confirm that no file appears in two partitions, and the manifest records the SHA-256 digest of the assignment file.

Four downloaded files proved unreadable. Two were recovered by re-downloading; two are permanently unavailable at source (0.02 % of the corpus) and were excluded.

The final corpus is **9 566 images, partitioned** 6 695/1 435/1 436. The split is uniform and unstratified, drawn once under seed 42, so the balance of the corpus carries into the partitions by sampling: class shares lie between 23.9 % and 25.9 % in all three. The physical properties were checked afterwards. The medians of SNR, duration, peak frequency and bandwidth are 14.97/14.09/14.98, 1.25/1.25/1.10 s, 75.3/75.3/75.3 Hz and 4112/4254/4131 Hz: peak frequency is identical across the three, bandwidth agrees to 3.5 %, SNR to a little over 6 %. Duration is the exception, the test median sitting 12 % below the other two.

3.2 Model

3.2.1 Architecture

`GWGlitchSNN` is a four-block convolutional spiking network. Each block is a 5×5 convolution with stride 2 and padding 2 followed by a LIF layer. The spatial downsampling is carried by the stride rather than by pooling, which keeps it inside the convolution. Average pooling would put analog values on a path that carries only spikes; max pooling keeps them binary but adds a second operator between a LIF layer and the convolution that follows it. Table 3.1 gives the layer configuration at the working resolution.

Stage	Operation	Output shape	Notes
Input	direct encoding $\times$T	[T, B, 3, 112, 112]	analog frame repeated
Block 1	Conv 3 $\rightarrow$ 16, k5, s2, p2 + LIF	[T, B, 16, 56, 56]	exposed as `spike1`
Block 2	Conv 16 $\rightarrow$ 32, k5, s2, p2 + LIF	[T, B, 32, 28, 28]	exposed as `spike2`
Block 3	Conv 32 $\rightarrow$ 64, k5, s2, p2 + LIF	[T, B, 64, 14, 14]	exposed as `spike3`
Block 4	Conv 64 $\rightarrow$ 128, k5, s2, p2 + LIF	[T, B, 128, 7, 7]	exposed as `spike4`
Readout	Flatten (6 272) $\rightarrow$ Linear $\rightarrow$ LIF	[T, B, 4]	one per class, summed over T

Table 3.1: Layer configuration at the working resolution of 112 px. The flattened width is derived from the input size at construction (25 088 at 224 px, 6 272 at 112 px, 2 048 at 64 px).

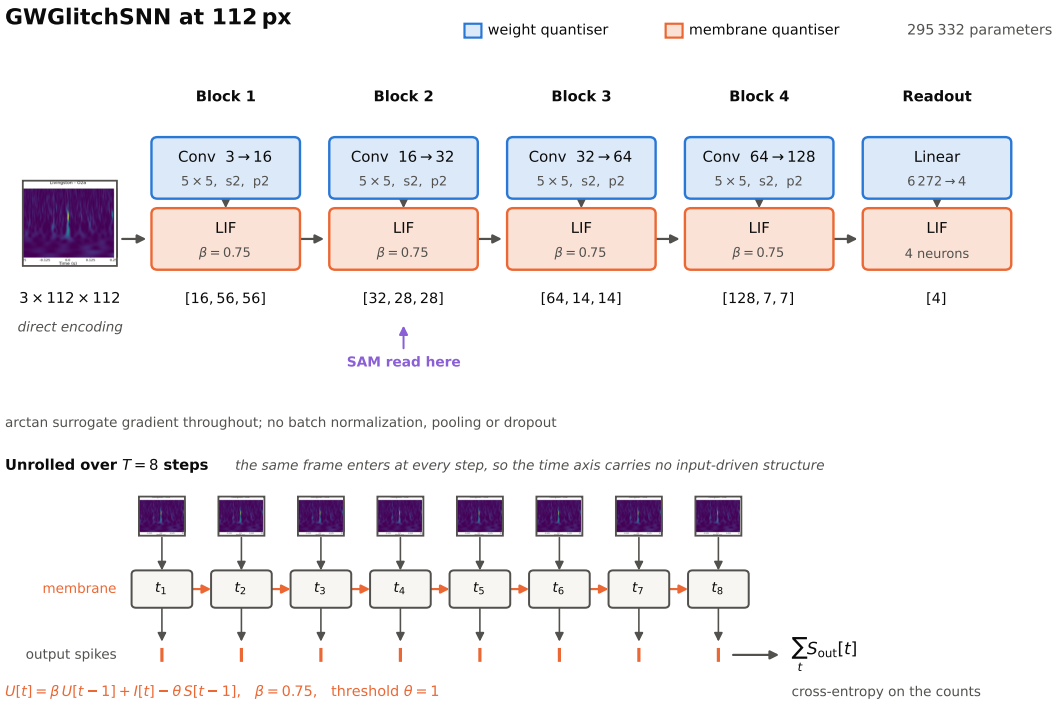


Figure 3.6: Datapath of `GWGlitchSNN` at the working resolution, and its unrolling over the T time steps.

The forward pass returns the stacked output spike train *and* the four hidden spike tensors of shape $[T, B, C, H, W]$, so that every diagnostic and explainability procedure consumes raw hidden spike trains directly from the model interface. Figure 3.6 draws the datapath and its unrolling over the T steps, the form in which those tensors are produced.

Parameter count is nearly resolution-independent: 295 332 at 112 px, of which 270 240 are convolutional and unaffected by input size, since only the linear readout shrinks with resolution, a fact with consequences for the design of the quantization grid discussed in Section 4.1.

Batch normalization, pooling and dropout are intentionally absent. Normalization layers interact non-trivially with quantization, and their treatment belongs to a later phase. The interaction is not only a quantization one: the statistics a spiking network would normalize accumulate over the time axis as well as over the batch, which is why two of the variants proposed for directly trained spiking networks redefine the operation — tdBN ties the normalization to the firing threshold [45], BNTT gives each time step its own learnable parameters [21]. On the hardware side two routes exist. The multiplication can be approximated by a shift, as in the shift-based normalization of binarized networks [19], or the layer can be removed altogether by folding its parameters into the preceding convolution before deployment [20].

3.2.2 Readout and loss

Classification uses a *rate-coded readout*: output spikes are summed over the T time steps and the resulting per-class counts are passed to a cross-entropy loss as logits. Optimization uses Adam [23]. This keeps SAM in the multi-spike regime, where its inter-spike-interval formulation remains meaningful.

3.3 Training

3.3.1 Hyperparameter optimization

Hyperparameters are searched with Optuna [5], maximizing validation accuracy. Each trial reports validation accuracy after every epoch and is pruned by a Hyperband pruner [25] (minimum resource 1 epoch, maximum 15, reduction factor 3), which allocates epochs to promising configurations and truncating the others early. Studies are persisted to SQLite, one study name per search and resolution.

Two studies were run. The first (v4), at 224 px resolution, searched the learning rate in the interval $[10^{-4}, 3 \times 10^{-3}]$ (log-uniform), β in $(0, 1)$ and $T \in \{5, \dots, 15\}$, and selected a learning rate of 1.036×10^{-4}, $\beta = 0.4148$ and $T = 7$. This was the configuration used for the resolution study. The second (v5), at the working resolution found after the study (112 px, as it will be better discussed in Section 3.4), imposed the deployment constraints described in Section 3.4.2 and produced the configuration used throughout the rest of the thesis.

3.3.2 Experiment tracking

Every trial is logged to Weights & Biases. Beyond validation loss and accuracy, the logged metrics also include the mean firing rate of each of the four hidden layers. Firing rate is the proxy for energy consumption in spiking hardware [36], so tracking it per layer during the search shows whether high-accuracy configurations achieve their accuracy through dense and expensive activity or through sparse spiking, and also whether any layer is silent or saturated before any explainability diagnostic is run.

A separately logged quantity, recorded under the name `avg_firing_rate`, is the mean over the four *output* neurons only. A value near 0.25 indicates that roughly one output neuron fires consistently, which is a signature of confident classification. It is relevant

to note that it is *not* the global firing rate of the network and is never used for energy accounting, for which only the per-hidden-layer rates are valid.

3.3.3 Final training

The training script reads the frozen configuration and trains the model. The script is parametrized on input resolution and seed, the same code indeed serves the resolution study, the final training and the multi-seed reference. The seed is set explicitly before the model is constructed. The checkpoint is written whenever validation accuracy improves, and training stops after five consecutive epochs without improvement in validation loss, which permits early stopping, and with a ceiling set at 50 epochs. Each run writes the checkpoint together with a record of the full per-epoch history, the per-layer firing rates, the per-class validation recall, the seed and the resolution.

3.4 Selecting the operating configuration

The final operating point is defined by the input resolution, the three hyperparameters, and the layer and decay constant at which the explanations are read. Each is settled on evidence measured on the validation partition and recorded in a single frozen manifest.

3.4.1 Input resolution

Nine training runs were performed: three resolutions (224, 112, 64 px) at three seeds each, on the frozen partition, with hyperparameters held fixed and only the input size varying. Since those hyperparameters had been found at 224 px, the protocol systematically handicaps the lower resolutions, so any deficit they show is an upper bound on the deficit attributable to resolution itself.

The procedure and its outcome are reported in Section 5.1.

3.4.2 Hyperparameter search under deployment constraints

The search uses Optuna [5] with a Hyperband pruner [25], maximizing validation accuracy over three hyperparameters chosen because they govern three distinct axes: learning rate (optimization stability), β (membrane leak and temporal memory), and T (latency, computation and memory cost).

Two deployment constraints are imposed inside the search space. The first restricts β to $\{0.25, 0.5, 0.75, 0.875\}$, values whose decay is realizable as a bit shift, the set spanning both families ($\beta = 2^{-k}$ and $\beta = 1 - 2^{-k}$). The second caps T at 8. Constraining the search guarantees that whatever it returns is deployable by construction, and Section 5.2 shows that this is not equal to rounding an unconstrained optimum afterwards: the free search selects $\beta = 0.4148$, whose admissible neighbours are 0.25 and 0.5, while the constrained search selects 0.75.

The lower bound of the learning rate was moved from 10^{-4} to 10^{-5}, because the v4 optimum (1.036×10^{-4}) sat on the boundary of the earlier range, and a boundary optimum should be investigated further to clarify whether it is a consequence of the decided range.

3.4.3 Baseline manifest

The outcome of these choices is recorded in a single file, `baseline_manifest.json`. It contains the input resolution, the three hyperparameters, the originating study, the decay rate and reference layer of the explainer, the list of seeds (1234, 2345, 3456), the checkpoint paths, the sizes of the training and validation partitions, and the first sixteen hexadecimal digits of the SHA-256 digest of the partition file (`754b2515b6509075`). Every downstream script reads the model and its configuration from the manifest.

3.5 Diagnostics of the network dynamics

Two diagnostics characterize whether a trained network is a healthy spiking system or a formally correct one whose neurons are dead or saturated. The two are reused unchanged on every quantized model.

Dead and saturated neurons. For each hidden layer, the mean firing rate of every individual neuron is accumulated across the validation partition, weighted by batch size. A rate below 0.01 identifies a *dead* neuron, while a rate above 0.80 a *saturated* one. Neither cutoff is derived, since thresholds of this kind are conventions read off the distribution: the pruning literature trims neurons whose fraction of zero activations exceeds one standard deviation of the layer mean [18], and the dormant-neuron analysis treats the cutoff as a tuned hyperparameter, reporting values between 0 and 0.1 [39].

The two ends are pathological for different reasons. A neuron firing on fewer than one input in a hundred is wasted capacity and, being silent, it cannot appear in any explanation; its output is near-deterministic, carrying 0.081 of the one bit a binary unit can hold. Saturation, instead, is not an information problem — a Bernoulli variable at 0.80 retains 0.722 bits so as much as one at 0.20 — but an energy one: the energy of a spiking layer scales with the number of spikes it emits [41], so a unit near its ceiling spends the sparsity the architecture was chosen for.

Regularity of firing. The coefficient of variation of the inter-spike intervals,

$$\mathrm{CV} = \frac{\sigma_{\mathrm{ISI}}}{\mu_{\mathrm{ISI}}}, \tag{3.1}$$

distinguishes regular, clock-like firing ($\mathrm{CV} \to 0$) from irregular, Poisson-like firing ($\mathrm{CV} \approx 1$). It is relevant here because the Spike Activation Map (SAM) is exactly built on inter-spike intervals. A random sample of 200 neuron positions per layer is fixed once at the first batch, and the intervals of exactly those positions are accumulated over every validation image. The CV is then computed only where at least five intervals were observed. Since a neuron can produce at most $T - 1 = 7$ intervals within a single image, pooling over the whole partition is what makes the estimate stable.

A second coefficient is computed inside the single image, over the intervals a neuron produces within the T steps of that image, and it assesses whether firing is periodic there. Two spikes give one interval whose standard deviation is zero by construction, so the per-image coefficient is read only where a neuron produced at least three spikes, and the full-precision parents are measured under the same rule.

3.6 The Spike Activation Map

The definition is the one given in Section 2.3: the recency weight of Equation 2.9, accumulated over a neuron's preceding spikes by Equation 2.10, masked by the spikes emitted at t and summed over channels by Equation 2.11. What the definition leaves open the implementation had to settle.

The summation range is exclusive of t. The score measures the contribution of a neuron's *preceding* trajectory, as including $t' = t$ would mean adding a term of weight $\exp(0) = 1$ for every neuron firing at t, so the map produced is the correct SAM *plus the raw binary spike map*. The pedestal is independent of both γ and the spike history, and therefore inflates every correlation between maps computed at different γ. A consequence of the correct definition is that SAM at $t = 0$ is identically zero, no past history existing at the first step.

The implementation uses the recursion

$$N_{t+1} = e^{-\gamma}\,(N_t + S_t)\,, \qquad N_0 = 0, \tag{3.2}$$

algebraically identical to Equation 2.10 but $O(T)$ rather than $O(T^2)$. A reference implementation computing the explicit sum is retained and the two are asserted equal at run time.

Visualizations overlay every time step on the denormalized input under a *shared colour scale* to assure that a change in intensity across time is genuine and is not an artefact of per-panel rescaling.

The reference configuration for every analysis below is **layer 2 with** $\gamma = 0.5$, both recorded in the manifest. The choice of γ is justified in Section 5.5.1.

3.7 Evaluating and comparing explanations

3.7.1 Multi-sample statistics

For each class, SAM is computed on 25 validation images, aggregated over time (mean over T) and stacked, and three maps are reported: the *mean*, meaning the region attended on average; the *standard deviation*, showing where the explanation varies most across instances; and their ratio, the *coefficient of variation*, which indicates whether a region is attended consistently or only in some examples. Mean and standard deviation share a colour scale across classes so that they are numerically comparable. The coefficient of variation has an extremely long tail where the mean is near zero, and its colour scale is truncated at the 95th percentile.

The same statistics are repeated for all four hidden layers, on 20 images per class. Since SAM sums over the channel axis and the four layers have 16, 32, 64 and 128 channels, raw magnitudes are not comparable across depth, and both maps are normalized by the channel count of their layer before being plotted on a shared scale.

3.7.2 Faithfulness: the deletion metric

Stability does not by itself imply correctness, so faithfulness is tested separately with the deletion metric of Petsiuk et al. [32]. Pixels are removed in decreasing order of attributed importance and the confidence of the model is read after each removal. A map that points at the evidence the model uses destroys that confidence faster than one that does not. Algorithm 1 states the procedure for a single image; `xai/deletion_metric.py` implements it.

SAM produces one map per time step and the metric needs one, so the maps are averaged over t, and what survives that average is an ordering of the pixels: only the ordering enters what follows. SAM is class-agnostic, so the tracked class is the one the model predicts on the unmodified image, held fixed while the pixels disappear. Removed pixels are set to zero in normalized space, which is the ImageNet channel mean of the preprocessing rather than black.

The pixel ranking is computed once per image and per ordering, and the mask at each fraction takes its first $\lfloor \phi HW \rfloor$ entries. The removed sets are nested: a pixel deleted at one fraction stays deleted at every larger one. What the curve traces is the progressive degradation of a single ranking, not 21 unrelated maskings.

At every fraction the two curves delete the same number of pixels and differ only in which ones. The gap between them isolates the ordering, the one thing the control changes, and cannot come from one condition removing more of the image than the other.

The area under a curve is the confidence the model retains, integrated over the fraction deleted: a faithful map destroys that confidence early and leaves less area behind than random deletion, which makes $\Delta = \mathrm{AUC}_{\mathrm{SAM}} - \mathrm{AUC}_{\mathrm{random}}$ negative, its magnitude the margin the map gains over chance. The 15 curves of a class are averaged point by

Algorithm 1 Deletion metric for one image

Require: network f, image x, reference layer ℓ, decay γ, fractions $F = \{0, 0.05, \dots, 1\}$
Ensure: confidence curves c^{SAM} and c^{rand} over F

$M \leftarrow \mathrm{SAM}(f, x, \ell, \gamma)$ {$[T, H, W]$, one map per time step}
$A \leftarrow \frac{1}{T} \sum_t M_t$
$A \leftarrow \mathrm{upsample}_{\mathrm{bilinear}}(A)$ {layer resolution to input resolution}
$\hat{y} \leftarrow \arg\max_c \mathrm{softmax}(f(x))_c$
for ord $\in$ {SAM, rand} **do**
  if ord $=$ SAM **then**
    $\pi \leftarrow \mathrm{argsort}(A,\ \mathrm{descending})$
  else
    $\pi \leftarrow$ uniform random permutation of $\{1, \dots, HW\}$
  end if
  for $\phi \in F$ **do**
    $x' \leftarrow x$ with pixels $\pi_{1 \dots \lfloor \phi HW \rfloor}$ set to 0 in all three channels
    $c_\phi^{\mathrm{ord}} \leftarrow \mathrm{softmax}(f(x'))_{\hat{y}}$
  end for
end for

point, and the area under the mean curve is taken by the trapezoidal rule, which joins consecutive samples by a straight segment and sums the 20 trapezia lying between the 21 fractions. The curve has no analytic form and only 21 empirical samples, so a higher-order rule would interpolate structure that was never measured.

3.7.3 Divergence metrics

Two explanations of the same image produced by two models are compared with different metrics addressing different questions. The first two read the stack summed over time, $A = \sum_t M_{\cdot,\cdot,t}$, and are spatial, while the third reads the stack unaggregated and is the only one that sees the time axis. Their provenance in the saliency literature is discussed in Section 2.3.

Spatial Pearson correlation (PCC). The linear correlation between the two summed maps, flattened over the HW positions,

$$\mathrm{PCC}(A, B) = \frac{\sum_i (A_i - \bar{A})(B_i - \bar{B})}{\sqrt{\sum_i (A_i - \bar{A})^2}\ \sqrt{\sum_i (B_i - \bar{B})^2}}. \tag{3.3}$$

Centring and dividing by the standard deviations makes it invariant to any affine rescaling $B \mapsto \alpha B + \beta$ with $\alpha > 0$, and that invariance is what allows a comparison across models whose overall spiking activity differs. Indeed, a quantized model that fires uniformly more than its parent is not charged for the offset, only for a change in the spatial pattern. The quantity is undefined when either map is constant, a case that in the implementation returns as missing rather than as zero.

Top-k intersection over union (IoU). Let A_k hold the positions whose value in A exceeds the $(1-k)$ empirical quantile of A, and B_k the same for B. The metric is the Jaccard index of the two sets,

$$\mathrm{IoU}_k = \frac{|A_k \cap B_k|}{|A_k \cup B_k|}, \tag{3.4}$$

with $k = 0.20$, the value used by Rabbi et al. [33]. Where the correlation weighs every position by its value, this reads the ranking alone and only at its top, as it asks whether the two models highlight the same region, not whether they assign it the same intensity. Two

independent rankings select kHW positions each and share k^2HW of them in expectation, over a union of $2kHW - k^2HW$, so the chance level is $k/(2-k) = 0.111$.

The threshold is a strict inequality against an empirical quantile, so the selected set holds exactly kHW positions only when the values are distinct. Ties can leave it empty, which happens when k is small enough for the quantile to fall inside a plateau, and the value is then undefined.

Temporal centre of mass (CoM). The mean time step at which the energy of the SAM stack is concentrated,

$$\mathrm{CoM} = \frac{\sum_t t\, E_t}{\sum_t E_t}, \qquad E_t = \sum_{i,j} M_{ij,t}, \tag{3.5}$$

with the steps indexed from 0 and $E_0 = 0$, SAM being identically zero at the first step. It is reported both as a signed difference between the two centres of mass, whose sign says whether the energy of the explanation moves earlier or later, and as its absolute value, which is the form the numerical floors are calibrated against. Both are in units of time steps. Aggregating over the whole map discards where the activity sits and keeps when it occurs, which is what makes it the complement of the two spatial metrics.

Beyond the three, the *per-timestep correlation* applies Equation 3.3 to the pair of maps at each t separately and returns a vector of length T in place of a scalar. It localizes the step at which agreement breaks down, which provides an information that neither a summed map nor a single centre of mass can show.

3.7.4 Calibration protocol

The divergence metrics are calibrated before they are used. One check precedes the calibration: a checkpoint compared with itself must return perfect agreement, meaning thaat two calls to the sampling routine must return bit-identical images and two forward passes of the same model must return identical maps. What follows are four reference divergences, measured on fifteen validation images per class at the reference layer and γ — twenty for the second, matching the grid it is read against.

1. **Numerical floor.** Gaussian noise is added to the weight tensors of one checkpoint, at magnitudes expressed as fractions of its measured 8-bit quantization step, and the perturbed model is then compared with the unperturbed one. Biases and the LIF buffers are excluded, since perturbing β would mix membrane dynamics with weight noise.

2. **Fine-tuning alone.** The control arm of the quantization grid (Section 4.1) runs the whole procedure with no bits removed. The result is the divergence that continued training costs on its own, and it is the operative threshold: a grid result closer to its parent than the control arm is a result about continued training, and not about the removed bits. Section 5.6.4 reports it.

3. γ **reference.** The sensitivity analysis gives the divergence produced by changing the explainer itself thirtyfold.

4. **Independent solutions.** The three full-precision checkpoints are compared pairwise in order to measure how similar two independently trained solutions are.

Two further reference points come from outside that protocol. The γ sensitivity analysis gives the divergence produced by changing the explainer itself thirtyfold. The control arm of the quantization grid (Section 4.1) gives the divergence produced by running the whole fine-tuning procedure with no bits removed, on twenty images per class rather than fifteen, matching the grid it is read against. The latter is the operative threshold: a grid result closer to its parent than the control arm is a result about continued training and not about the removed bits; section 5.6.4 reports it.

Chapter 4
The Quantization Study

This chapter describes how the quantized models are built and how they are compared with their full-precision parents. Every design choice below is either inherited from the calibration of Section 5.6 or settled by a measurement reported in Section 5.7. Again, the whole analysis is performed on the validation partition.

4.1 Design principle: three separate arms

The fundamental design choice is to **quantize the two targets separately before quantizing them together**, as separating them turns the grid into a causal ablation and is what makes the hypothesis of Chapter 1 testable. Table 4.1 lists the configurations that follow; Figure 4.1 shows where in the datapath each arm acts, and that the two never act on the same tensor.

Arm	Weights	Membrane potential	Purpose
FP32 reference	32-bit float	32-bit float	baseline for every comparison
FP32 control	32-bit float	32-bit float	separates the pipeline from the bits
Weight-only (W)	8 / 4 / 2 bit	32-bit float	isolates the synaptic (spatial) effect
Membrane-only (U)	32-bit float	8 / 4 / 2 bit	isolates the state (temporal) effect
Joint (W+U)	8 / 4 / 2 bit	8 / 4 / 2 bit	deployable configuration

Table 4.1: The quantization grid: nine quantized configurations and one control, each at the three seeds of the reference (30 fine-tuning runs).

The *control* arm runs the whole fine-tuning procedure with no quantizer attached. Both bit-widths are left unset, so neither the weight parametrization nor the membrane hook is ever registered and the network trains in full precision. Its purpose is subtractive. The nine quantized configurations differ from their parent in two respects at once, the bits removed and the operations that remove them — reloading the parent, fine-tuning it, selecting a checkpoint, re-evaluating — and the control runs those operations alone, so whatever it produces is their contribution and theirs alone.

It produces two sets of weights. The first is the checkpoint the selection rule returns, and it is bit-identical to the parent: the same label on all 1 435 validation and 1 436 test images, per-layer firing rates equal to the last digit reported. The identity follows from the rule: the selection begins with the un-fine-tuned model already held as a candidate and replaces it only when an epoch exceeds its validation accuracy *strictly*. With nothing quantized, that initial candidate sits at the parent's own accuracy, so an epoch would have to beat the parent to displace it, and on none of the three seeds did it. What the control establishes is that reloading the parent, selecting a checkpoint and re-evaluating leave the

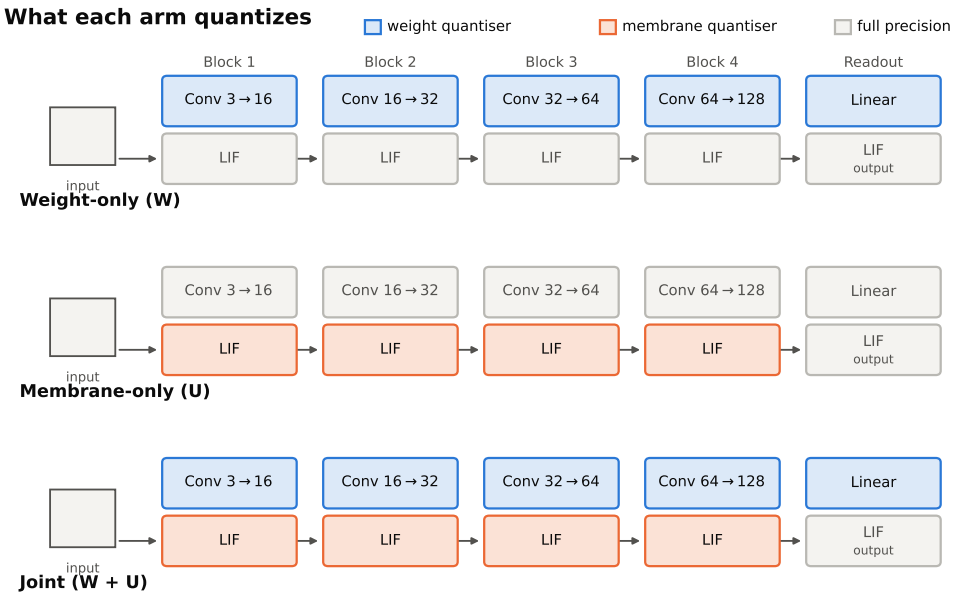


Figure 4.1: The three arms, as which parts of the datapath each one quantizes.

model bit-identical, so none of the divergence measured in the other nine configurations can be charged to the apparatus around the training. The training itself is not covered by that argument, and it is the subject of the second set of weights.

The second is the state of the weights after the last epoch, kept separately and not identical to the parent. Fine-tuning moved them even though it never succeeded in replacing the checkpoint. What that state measures is the effect of continued training alone on an explanation, and it is the fine-tuning level of the calibration (Section 5.6.4).

Two quantities are deliberately left out of the grid. Firstly the membrane decay $\beta = 0.75$, which is already shift-implementable, having been constrained inside the hyperparameter search (Section 3.4.2) and is not re-quantized. Secondly, the biases are kept in full precision: there are 244 of them, under 0.1 % of the parameters, and on integer hardware they are added once to a wide accumulator.

The two arms respond differently to the resolution choice. 270 240 of the 295 332 parameters are convolutional and therefore resolution-independent, so what the weight arm saves is nearly constant across resolutions. The membrane state, which is what the other arm quantizes, falls twelvefold between 224 and 64 pixels.

4.2 Quantization-aware fine-tuning from the parent

Quantization-aware training is implemented by fake quantization: in the forward pass each quantized tensor is rounded to its grid and then clamped to the grid's bounds, and in the backward pass the rounding is replaced by the identity, the straight-through estimator [7].

The two quantizers are attached by different means because they act on different kinds of object: a weight is a parameter, owned and updated by the optimizer, while a membrane potential is a state that is rebuilt at every time step.

Weights are quantized through a *parametrization*. PyTorch moves the stored tensor aside and recomputes `weight` from it on every access, so the module's weight becomes a function of a hidden full-precision tensor. The optimizer holds and updates that hidden tensor, the forward pass only ever reads its quantized image, and the straight-through estimator carries the gradient back to it unchanged. Keeping the full-precision copy is essential to

fine-tuning and makes it possible: an update smaller than half a quantization step applied to a value already on the grid would round away and leave nothing behind, whereas if accumulated in the hidden tensor many such updates eventually carry the weight to the next level.

When the training ends the parametrization is folded away and the quantized values are written into ordinary parameters, so the saved checkpoint holds weights already sitting on the grid.

The membrane is quantized through a *forward hook* on each of the four hidden LIF layers. A hook that returns a value replaces the output of the module it is attached to, and a `Leaky` layer returns the pair of spike and membrane: the hook passes the spike untouched, binary as it already is, and substitutes the quantized membrane. That substituted value is the one the forward loop carries into the following step, so the grid constrains the state as it crosses a time step and reaches the spikes only by way of the step after. The membrane grid is fixed in advance, unsigned on $[0, 2]$ from the clipping study.

Every quantized model is fine-tuned from the full-precision checkpoint of the same seed. This is the constraint that guarantees that the quantization effect is distinct from initialization variance. The numerical floor of the calibration (PCC $= 0.975$) holds for comparisons between a model and its own parent, where no channel permutation exists, so a quantized model trained from a random initialization would be an independent solution, for which the relevant level is 0.153 and almost no quantization effect would be distinguishable from initialization variance.

Schedule. The fine-tuning is short and at reduced learning rate, adapting the parent to its grid: Adam is set at one tenth of the reference learning rate (1.047×10^{-5}), at most 20 epochs, early stopping occurs after 5 epochs without improvement of the validation loss.

Before the first epoch the model is evaluated with the quantizer attached and no training, which gives the *post-training quantization* (PTQ) accuracy of the same configuration at no extra cost: that un-fine-tuned model is saved as it represents the first candidate checkpoint, and it is replaced only by an epoch that strictly improves the validation accuracy. The selected checkpoint is the one on which every subsequent measurement — accuracy, explanations, firing statistics, energy — is taken. The 30 runs took 13 hours on the same 6 GB consumer GPU used for every training run in this thesis.

Gates before the grid. Two checks were run before any fine-tuning to validate the quantizer and then to validate the PTQ gate. The first verifies the quantizer itself: attaching and removing it restores the original weights exactly, and folding the quantized values into a plain checkpoint and reloading it leaves no drift (0 at 4 bits).

The second targets the PTQ gate: the joint 8-bit model, with no fine-tuning, must indeed match its parent to within the spread of the full-precision reference, four samples out of 1 435. At 8 bits the weight step is of the order of the perturbation that left the spatial correlation at 0.975 in the calibration, so an appreciable loss of accuracy there would effectively indicate a defect of the implementation. Both gates passed (Section 5.9.1).

All 8-bit configurations were run first, then 4-bit, then 2-bit.

4.3 Weight quantizer

Weights are quantized on a *symmetric signed* grid whose outermost level is $q_{max} = 2^{b-1} - 1$ on each side of zero, for a total of $2q_{max} + 1$: 255 levels at 8 bits, 15 at 4 bits and 3 at 2 bits (the values $-\Delta$, 0, $+\Delta$). Zero is exactly representable at every bit-width. The quantizer is applied to every convolutional and linear weight tensor, including the readout.

The scale is computed per output channel. A single scale for the whole tensor is set by its largest weight, so a channel whose own weights are small reaches only the innermost levels of the grid and leaves the external ones unused. The measured distributions of the reference checkpoint make this effect concrete (Table 4.2): indeed in the deeper convolutions the largest weight of the weakest channel falls below half the largest weight of the tensor. This halves the levels that channel can reach and costs it one bit, which at 2 bits it is critical: the signed grid is $\{-\Delta, 0, +\Delta\}$ with Δ the tensor maximum, so every weight of a channel whose maximum lies below $\Delta/2$ rounds to zero and the channel vanishes. A per-channel scale takes Δ from each channel's own maximum, so every channel reaches the outermost level whatever its magnitude. Figure 4.2 draws the two cases for the weakest channel of conv4 at 4 bits: seven of the fifteen levels are reachable under a scale set by the tensor, fifteen of fifteen under the channel's own.

Layer	max $\lvert w \rvert$, tensor	max $\lvert w \rvert$, weakest channel	Ratio	Bits lost per-tensor
conv1	0.144	0.115	1.25	0.32
conv2	0.126	0.060	2.10	1.07
conv3	0.113	0.054	2.09	1.07
conv4	0.099	0.044	2.25	1.17
linear	0.214	0.146	1.47	0.55

Table 4.2: Weight ranges of the reference checkpoint (seed 1234). "Bits lost" is $\log_2$ of the tensor maximum divided by the weakest channel's maximum: the resolution that channel ceases under a single per-tensor scale.

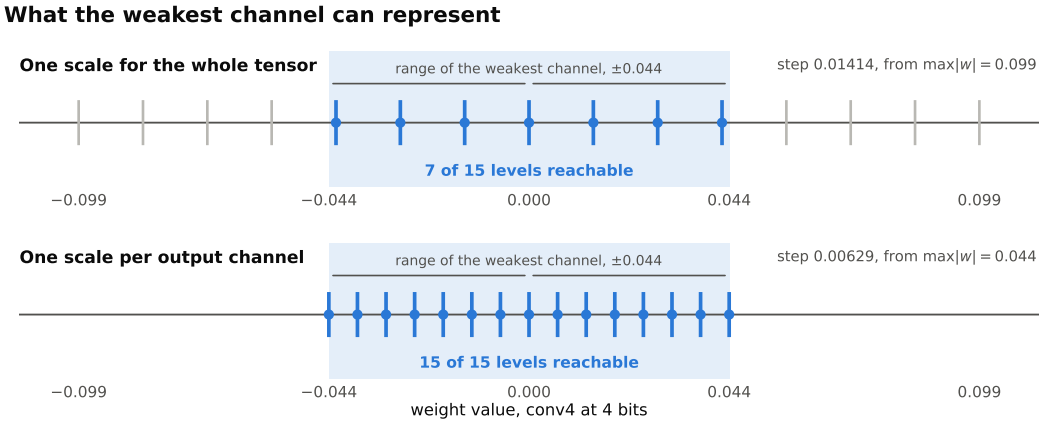


Figure 4.2: Grid levels reachable by the weakest channel of conv4 at 4 bits, under a scale derived from the tensor maximum and from the channel's own.

The global 8-bit step used in the calibration (0.00168, from the largest absolute weight of the network, 0.214) is therefore an upper bound on the per-channel steps actually applied, whose mean is between 0.0005 (conv4) and 0.0013 (readout) at 8 bits.

4.4 Membrane quantizer

The membrane potential of the four hidden LIF layers is quantized at every time step. The quantizer acts on the potential returned by each layer after the spike decision, so the quantized value is the *state* carried into the next step: what is stored, and therefore what costs memory, is what is quantized. The membrane of the output layer (four neurons) is left in full precision because its state is negligible in memory and quantizing it would

act on the spike decisions of those four neurons, whose summed counts are the logits; its *weights* instead are quantized with every other linear tensor (Section 4.3), so the readout is split between the two arms.

As has been established, a grid for the membrane can not be designed without knowing the range of the quantity, and unlike a weight, the potential has no range until the network is run. Its distribution was therefore measured over the whole validation partition (Section 5.7), and the clipping range was chosen by testing candidate ranges on the full-precision network, so before any quantization.

The weight scheme is the symmetric per-channel formulation of Jacob et al. [20]. For the membrane no comparable convention exists: MINT [43] quantizes the potential of a spiking network, but ties its scale to the weights, which is the coupling Section 4.5 avoids.

The membrane grid is unsigned over $[0, 2]$, twice the firing threshold $U_{\text{thr}} = 1$, with 2^b levels starting exactly at zero. The step is $2/(2^b - 1)$:

Bits	Levels	Step	Step / threshold
8	256	0.0078	0.8 %
4	16	0.1333	13.3 %
2	4	0.6667	66.7 %

Table 4.3: The membrane grid at each bit-width. At 2 bits the representable potentials are 0, 2/3, 4/3 and 2: a single level lies strictly between rest and threshold.

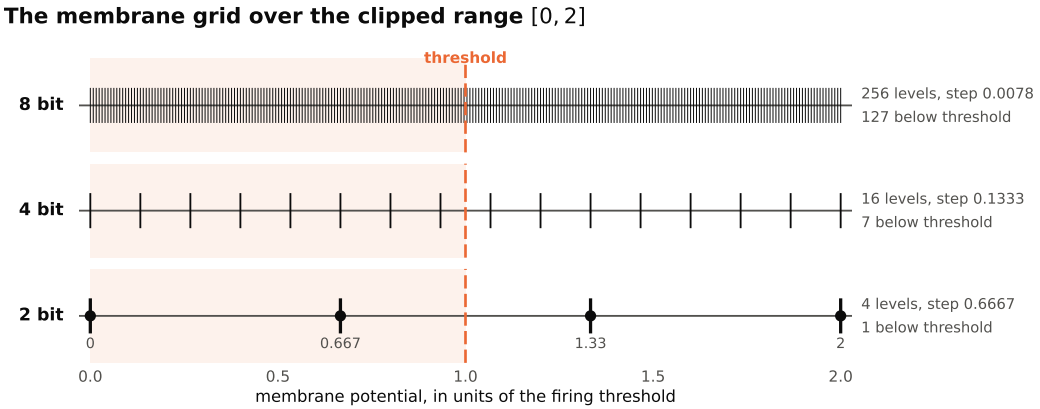


Figure 4.3: The membrane grid at each bit-width over the clipped range, with the firing threshold marked.

Zero is the value to which every neuron is initialized and around which it rests, so it must be exactly representable, which the unsigned grid guarantees by construction, its first level being zero.

Table 4.3 contains a quantitative prediction. The step is 0.8 % of the threshold at 8 bits, 13.3 % at 4 bits and 66.7 % at 2 bits, where it is of the same order as the threshold itself, and where a neuron's state can encode only whether it is at rest, part-way to threshold, or above it. Figure 4.3 draws the three grids on one axis: the 8-bit row reads as a ruler, the 2-bit row as four states. This is the mechanism behind the observation of Smith et al. [38] that membrane quantization needs at least four bits, and the grid tests it directly.

The unsigned grid follows from the clipping study. While potentials do go negative: the median of `lif1` sits at -0.08 and its first percentile at -6.6, clipping them away changes nothing the classifier uses. The symmetric range $[-2, +2]$ and the unsigned range $[0, +2]$ return the same accuracy to the last digit, 0.9826, one validation sample below the unclipped network, so a signed grid would spend half of its levels below zero, where no

decision is made. At 8 and 4 bits that costs a factor of two in resolution, one bit. At 2 bits it costs a factor of three: the step grows from 2/3 to the full 2, and no level at all remains between rest and threshold.

4.5 Joint arm

The joint arm applies both quantizers at once, at the same bit-width, with the per-channel weight scales of Section 4.3 and the fixed membrane grid of Section 4.4. **The two scales are independent.** MINT [43] shares a single scale between weights and membrane so that the inference loop needs no rescaling multiplier, but that choice is an optimization of a specific integer datapath, and imposing it here would couple the two targets that the grid exists to separate. With independent scales the joint arm is mechanically the composition of the two single-target arms: the two quantisers never see each other's grids or state, and that any outcome of the joint arm that the two single-target arms do not account for is therefore an interaction within the network, not a coupling introduced by the instrument.

4.6 Reading the results

For every quantized model, SAM maps are recomputed at the reference layer and γ on twenty validation images per class, and compared with those of **the full-precision parent of the same seed**, using the metrics of Section 3.7.3: spatial PCC, top-20 % IoU, the temporal centre of mass (absolute and signed shift), and the per-timestep correlation curve.

Further measurements characterize the behaviour:

- **cross-faithfulness**: the deletion metric applied to the quantized model with its *own* map and with the map of its *parent*, on fifteen validation images per class and twenty-one deletion fractions. The deletion baseline is recomputed for the parent of each seed, and a class is interpretable for a seed only if its baseline gap between SAM and random deletion is at least 0.05;
- the **diagnostics** of Section 3.5, collected for every model in a single pass over the validation partition: per-layer firing rate, dead and saturated fractions, CV-ISI, and per-sample predictions;
- the **Earth Mover's Distance** between the per-neuron firing-rate distributions of quantized model and parent, for a direct comparison with Smith et al. [38].

The CV-ISI follows the protocol of Section 5.3 — 200 positions per layer, seed 42, a coefficient computed only where at least five intervals are available.

Five principles, established in Chapter 5 before the grid was run, aid in the reading of those numbers.

1. **Below the numerical floor is noise.** A divergence smaller than the one produced by weight noise of one 8-bit step (PCC 0.975, IoU 0.704, CoM difference 0.029 steps, worst class) can not be deemed a result. The threshold that decides whether a result belongs to the removed bits sits above it, at the fine-tuning level of Section 5.6.4.
2. **IoU is an early indicator, and PCC a graded measure.** IoU has a dynamic range of 2.6× its floor and saturates early, while PCC has 34×.
3. **Everything is reported per class and per seed.** The sensitivity ordering Violin_Mode > Scattered_Light > Blip > Extremely_Loud recurs under every intervention tested, that is why a pooled figure would average a class that responds strongly with one that does not.

4. **A flat result on Extremely_Loud is uninformative.** The calibration moved that class by nothing: PCC 0.9997 under the largest perturbation applied, worst per-timestep correlation is 0.996, a sensitivity in $1 - \text{PCC}$ 94 times below Violin_Mode's. On the other three classes the calibration showed the metrics responding to a change known to have been made,but Extremely_Loud never produced a response to calibrate against. This means that a flat reading is consistent both with an explanation that did not move and with an instrument that could not have seen it move.

5. **Cross-faithfulness is interpretable only where the parent has a baseline signal.** On the reference seed that means Blip and Extremely_Loud; since the baseline is recomputed per seed, the set of interpretable classes is itself reported.

Every quantized model is compared with *its own* parent, and never with the parent of another seed, which is the principle that keeps together the other ones and the whole analysis. Because the three parents differ from each other far more than any quantized model differs from its own (Section 5.6), a comparison against the wrong seed would produce a large spurious divergence. The guard against that error is that in every comparison the 8-bit weight arm, the smallest intervention in the grid, must keep a spatial correlation above 0.98 with its parent in every seed.

The test of the hypothesis. As discussed, the hypothesis predicts an *ordering*: the membrane arm should displace the temporal centre of mass beyond its floor at a higher bit-width than the one at which the weight arm lowers the spatial correlation beyond its floor. The reverse order, or a simultaneous change falsifies it.

4.7 Stratified analysis

Joining the per-sample outputs to the physical descriptors allows the result to be reported as a relation, which an aggregate divergence does not permit.

The hypothesis is that divergence increases as event strength decreases, with a mechanism specific to spiking networks: quantizing the membrane potential flips a neuron's spike decision only where the potential lies within the quantization error of the threshold, and weak events are exactly what produce potentials that sit close to threshold. Because a flipped spike is a discrete event that propagates both to the next layer and forward in time through the decay and reset, and since SAM is constructed from spike trains, the flip changes the object the explanation is made of.

The prediction discriminates between arms: membrane quantization applies a fixed absolute error and should show a steep dependence on event strength, whereas weight quantization perturbs the input current approximately in proportion to the signal and should be flatter. It is tested on the membrane arm at 4 and 2 bits, the configurations in which the arm has an effect and accuracy is intact.

Two preconditions were verified. SNR does translate into image intensity, the Gravity Spy colour scale being fixed, not auto-scaled per image. SNR is also very nearly a proxy for class: on the validation partition the top SNR quartile is 89.7 % Extremely_Loud. Stratification uses **terciles computed within each class**, fifteen images per class and tercile. The summary statistic is the IoU gradient, high-SNR tercile minus low-SNR tercile; a positive gradient means that weak events lose their explanation more.

4.8 Energy and memory

Synaptic operations are counted per layer as

$$\text{SOP}_\ell = \text{f}_{\ell-1} \cdot \text{T} \cdot \text{MAC}_\ell, \tag{4.1}$$

where $f_{\ell-1}$ is the mean firing rate of the preceding hidden layer and MAC_ℓ the multiply-accumulate count of the equivalent dense layer, and multiplied by the per-operation costs of [17]. The rates are the per-hidden-layer figures measured on the *selected checkpoint* of each configuration, in the same validation pass that produces its diagnostics, so that energy, accuracy and explanations all refer to the same model.

One layer is asymmetric. Under direct encoding the first convolution receives analog pixel values rather than spikes, therefore its operations are multiply-accumulates. Because the input is identical at every time step, the current it produces is constant in time and is computed once rather than T times: it is a standard implementation of constant encoding that avoids recomputing that layer at every step, which would make the spiking network more expensive than its convolutional equivalent (Section 5.8).

Per-operation costs. Full-precision configurations, the membrane arm included, use 4.6 pJ per MAC and 0.9 pJ per accumulate; configurations with quantized weights use the 8-bit integer figures, 0.2 and 0.03 pJ. Narrower integer operations are not tabulated, so at 4 and 2 bits the costs are extrapolated: a multiplication scales quadratically with the operand width and an accumulate linearly, giving 0.05 and 0.015 pJ at 4 bits and 0.0125 and 0.0075 pJ at 2 bits. These two rows are estimates.

The first convolution is costed at the weight bit-width, which assumes that the input image is quantized to the same width; Section 5.9.7 shows how much the conclusion depends on that assumption.

Memory is the number of bits needed to store the 295 088 weights plus those needed to store the membrane state of the 94 080 hidden neurons at the working resolution.

Chapter 5
Results

5.1 Input resolution

5.1.1 Accuracy does not discriminate

Input size	Val. accuracy	Std. dev.	Observed range	Epochs
224 px	0.9854	0.0007	0.9847–0.9861	22.7
112 px	0.9800	0.0035	0.9777–0.9840	17.3
64 px	0.9814	0.0011	0.9805–0.9826	24.7

Table 5.1: Validation accuracy by input resolution, mean and standard deviation over three seeds.

The ordering in Table 5.1 is not monotonic — 64 px outperforms 112 px — and a genuine resolution effect would produce monotonic degradation. The arithmetic agrees: 0.9854 corresponds to 21.0 errors out of 1 435 validation samples and 0.9814 to 26.6, a gap of 5.6 samples, against a 95 % Wilson interval on a single measurement of ± 0.7 percentage points. The four classes survive an elevenfold reduction in pixel count, consistent with their being distinguished by coarse morphology rather than fine texture.

Per-class recall is directional but small. Violin_Mode degrades most from 224 px to 64 px, by 1.00 percentage points, which is the direction predicted from morphology since narrow spectral lines are the structure most vulnerable to downsampling — but the magnitude is 3.6 samples out of 365, inside the confidence interval of a single measurement. Extremely_Loud degrades more at 112 px than at 64 px, a second non-monotonicity. **No class collapses at reduced resolution.**

5.1.2 Network health does discriminate

Since accuracy is uninformative, the resolution was decided on the internal state of the network. For each hidden layer, the mean firing rate of every individual neuron is accumulated across the validation partition. A rate below 0.01 identifies a *dead* neuron, above 0.80 a *saturated* one.

The two ends are pathological for different reasons. A neuron firing on fewer than one input in a hundred is wasted capacity and, being silent, cannot appear in any explanation; its output is near-deterministic, carrying 0.081 of the one bit a binary unit can hold. Saturation is not an information problem — a Bernoulli variable at 0.80 retains 0.722 bits, as much as one at 0.20 — but an energy one: the energy of a spiking layer scales with the number of spikes it emits [41], so a unit near its ceiling spends the sparsity the architecture was chosen for.

Neither cutoff is derived. Thresholds of this kind are conventions elsewhere too: the pruning literature trims neurons whose fraction of zero activations exceeds one standard deviation of the layer mean [18], and the dormant-neuron analysis treats the cutoff as a tuned hyperparameter, reporting values between 0 and 0.1 [39]. Sweeping both cutoffs on the adopted configuration shows no plateau (Figure 5.1). Half an order of magnitude either side of 0.01 moves the dead fraction by 11.2, 9.0, 6.7 and 11.2 percentage points in the four layers, so the distribution of rates holds no gap separating dead units from live ones. The dead fraction is the value of a monotone function read at a conventional point rather than the size of a well-defined population, and it carries meaning only as a comparison at fixed cutoff, which is how Table 5.2 uses it.

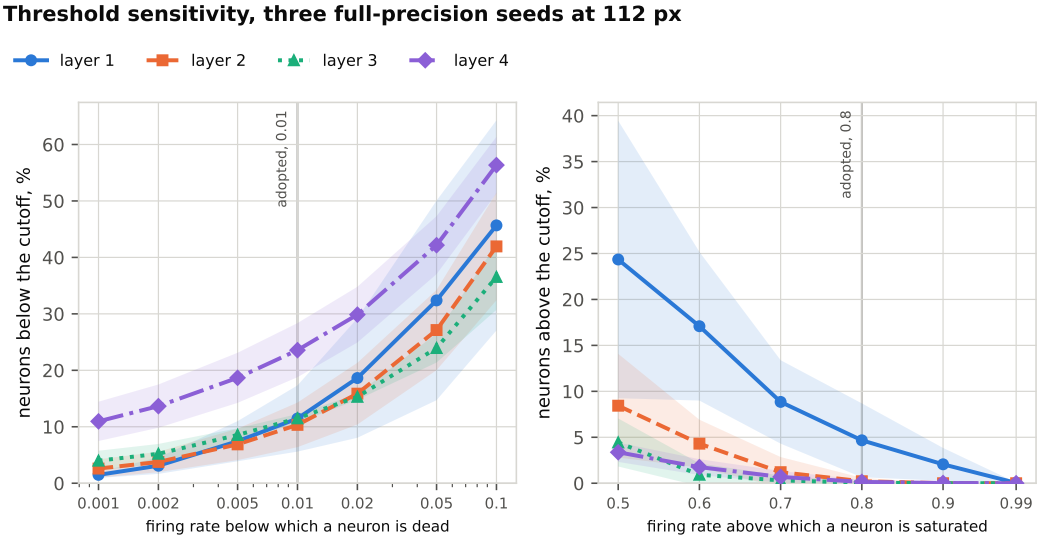


Figure 5.1: Dead and saturated fractions against the cutoff that defines them, mean and standard deviation over the three full-precision seeds at 112 px.

Input size	Layer 1	Layer 2	Layer 3	Layer 4
224 px	12.1 ± 3.3	16.6 ± 4.9	12.6 ± 2.3	$\mathbf{36.3 \pm 4.0}$
112 px	17.8 ± 3.4	11.8 ± 0.8	13.0 ± 1.0	$\mathbf{23.6 \pm 2.9}$
64 px	16.4 ± 4.8	11.2 ± 2.3	11.7 ± 2.3	$\mathbf{19.5 \pm 1.5}$

Table 5.2: Percentage of dead neurons by layer and resolution, mean $\pm$ standard deviation over three seeds, on validation.

Layer 4 is the only layer that responds to resolution, and it responds on two measures at once: the dead fraction falls monotonically from 36.3 % to 19.5 %, while the median per-neuron firing rate rises from 0.0399 to 0.1499. These were the two competing explanations for the higher mean activity observed at lower resolution — more neurons participating, or the same neurons firing harder — and both occur. Saturation does not increase to compensate: neurons above 0.80 remain below 2.5 % in layer 4 at every resolution.

The interpretation is spatial redundancy, not a cured pathology. Layer 4 contains 25 088 neurons at 224 px and 6 272 at 112 px, so in absolute terms the number of *live* neurons falls from roughly 15 970 to 4 789. Reducing the resolution does not increase usable capacity; it reduces capacity while preferentially discarding units that were contributing nothing. At 224 px the deepest layer views a 14×14 map derived from a largely empty spectrogram, and a third of its units never encounter signal.

5.1.3 Decision: 112 pixels

Most of the benefit comes from the first halving: 224 to 112 px recovers 12.7 percentage points of dead fraction, the further move to 64 px only 4.1 more. Membrane state at 8 bits falls from 368 KiB to 92 KiB, bringing the full inference state within the SRAM budget of the target class of devices.

The decisive argument against 64 px is the spatial resolution of the explanations themselves. SAM maps are computed on layer activations, whose side lengths are 56, 28, 14 and 7 at 112 px but 32, 16, 8 and 4 at 64 px. A 4×4 map contains sixteen values, on which centre-of-mass, top-K intersection-over-union and structural similarity are all meaningless. Since the contribution of this thesis is the fidelity of explanations rather than of predictions, a configuration in which the measured object is degenerate in one layer cannot be accepted whatever it saves in compute.

5.2 Hyperparameters and the full-precision reference

The constrained search selected $\beta = 0.75$, $T = 8$, **learning rate** 1.047×10^{-4} at a best validation accuracy of 0.9847 within the 15-epoch budget.

The constraint on β is justified empirically, not only by argument. An unconstrained search over $\beta \in (0, 1)$ selects 0.4148, whose nearest admissible neighbours are 0.25 and 0.5. The constrained search selects 0.75. Rounding the unconstrained optimum after the fact — the alternative to constraining the space — would therefore have produced a configuration that is both inferior and never validated.

All four values were sampled; the search allocated 40 % of trials to 0.75 and 10 % to 0.875, which is itself the surrogate model's judgement after early exploration. Among trials surviving at least three epochs, 0.75 leads on both the best (0.9847) and the median (0.9763) validation accuracy, while 0.875 trails on both and was pruned in all three of its trials.

$T = 8$ **is an optimum at the imposed ceiling** and is reported as such: the search was forbidden from exploring $T > 8$. The mitigating evidence is that an unconstrained search with range up to 15 selected $T = 7$, so the preference for long integration windows is mild.

5.2.1 β improves the measurement instrument

Holding resolution and partition fixed and changing only β from 0.4148 to 0.75, the layer-4 dead fraction is *unchanged* at 23.6 %. β does not affect the dead fraction; resolution does. This strengthens the reading of Section 5.1: the deep-layer problem is spatial redundancy rather than a dynamical pathology curable by a better decay constant.

What β does change is saturation, by an order of magnitude and most where it matters. Neurons above 0.80 fall from 9.3/3.5/2.7/1.8 % across the four layers to 4.7/0.22/0.06/0.14 %. The layer-2 figure is the consequential one, because a saturated neuron accumulates the maximum possible contribution score on every image and therefore contributes a constant, non-discriminative floor to every SAM map. **Selecting β on validation accuracy alone happened to reduce that contamination by a factor of sixteen at the reference layer.**

5.2.2 Three-seed reference

The reference is statistically indistinguishable from the 224 px configuration (0.9854): three samples out of 1 435 separate them, against a four-sample spread across seeds. **Reducing the pixel count by a factor of four, and constraining β and T to hardware-realizable values, costs no measurable accuracy.**

Accuracy is reproducible; internal dynamics are not. At fixed accuracy the mean firing rate of layer 1 varies by a factor of 2.6 across seeds, and the coefficient of varia-

Quantity	Mean	Std
Validation accuracy	0.9833	0.0014
Epochs to early stop	21.0	1.7
Recall, Blip	0.9915	0.0028
Recall, Extremely_Loud	0.9855	0.0000
Recall, Scattered_Light	0.9839	0.0027
Recall, Violin_Mode	0.9726	0.0099

Table 5.3: Full-precision reference at 112 px, $\beta = 0.75$, $T = 8$, over three seeds on validation.

tion falls monotonically with depth: 46.9 % at layer 1, 28.6 % at layer 2, 15.3 % at layer 3, 9.5 % at layer 4. The regime is established within the first epoch and persists, indicating that initialization selects the firing regime and training does not subsequently equalize it. A measurement taken during the resolution study puts the effect in the units of the closest work: the Wasserstein-1 distance between the per-layer firing-rate profiles of two full-precision models differing only in seed is 0.039, inside the range of Earth Mover's Distances that Smith et al. [38] report *between full-precision and quantized* models. This is the observation that motivates Section 5.6.

5.3 Physical validation of the dynamics

At the adopted configuration the confusion matrix diagonal reads 0.99/0.99/0.98/0.97, consistent with Table 5.3; the only non-negligible errors are on Violin_Mode, the class that in the tabular domain partially overlaps both Blip and Scattered_Light (Figure 5.2).

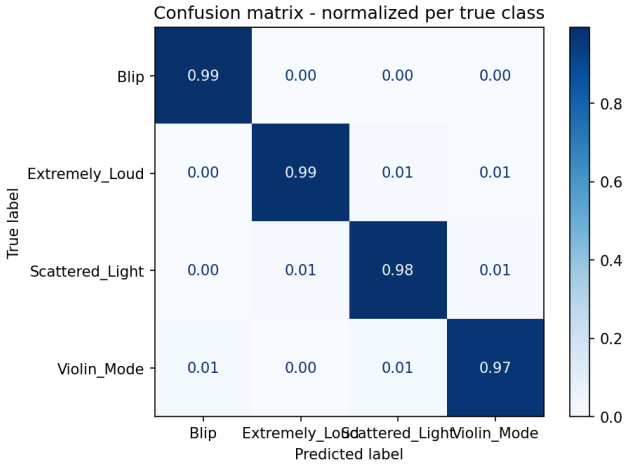


Figure 5.2: Confusion matrix of the reference checkpoint (seed 1234) on validation, normalized per true class.

The dead fraction is 16.1/12.6/11.3/22.8 % across the four layers (Figure 5.3). Saturation is confined to the first layer and to 24 of its 50 176 neurons, 0.05 %, with none above the threshold in the other three. The reference checkpoint is the quiet one. Across the three seeds the layer-1 saturated fraction is 4.67 ± 4.02 %, and in two of them neurons fire on more than 99 % of steps: 13 at seed 2345, 40 at seed 3456. Layer 1 moves in both directions at once, holding the fewest dead units in the seed where it holds the most saturated ones. These figures reproduce those of Section 5.1 exactly for the same seed, computed through an independent code path, which is a free verification of both.

One feature deserves note: the layer-1 distribution is **bimodal**, with a secondary population near a firing rate of 0.65–0.75 that has no counterpart in the deeper layers. A high-activity subpopulation therefore exists in the first layer only, which may be related to its being the layer with the largest coefficient of variation across seeds.

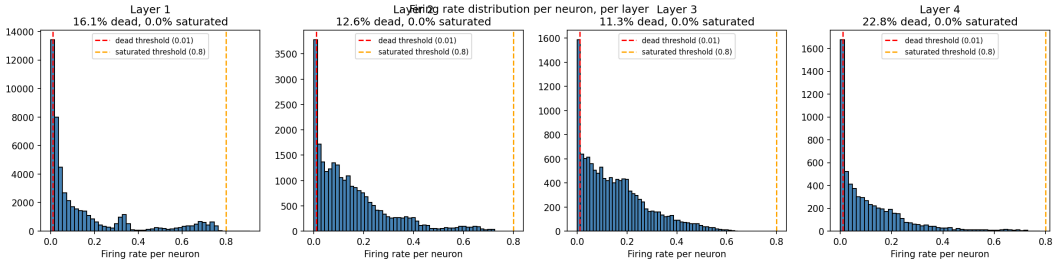


Figure 5.3: Distribution of per-neuron firing rates in the four hidden layers, on validation, with the dead (0.01) and saturation (0.80) thresholds.

The coefficient of variation of inter-spike intervals concentrates between 0.35 and 0.45 in all four layers. Firing is regular rather than Poisson-like, which is the expected outcome under constant direct encoding: since the input does not vary over time, any irregularity observed is generated entirely by the membrane dynamics and by interaction between layers. This is evidence about internal dynamics, not a claim of biological realism. With $T = 8$ a neuron can produce at most seven intervals, so the five-interval threshold used for the estimate is close to the ceiling, which is why the accumulation runs over the whole partition rather than a single image. Between 184 and 196 of the 200 sampled neurons per layer met the threshold (Figure 5.4).

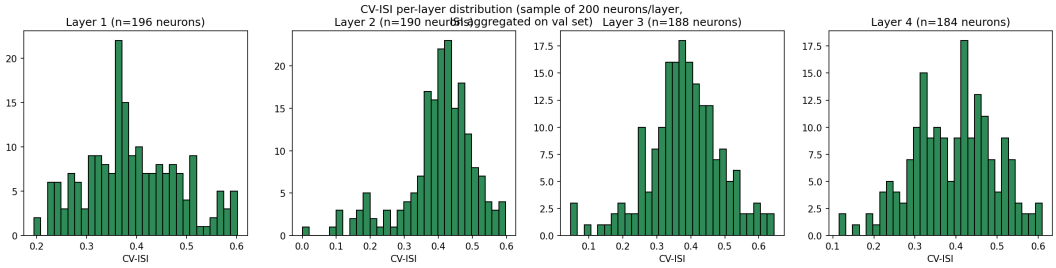


Figure 5.4: CV of inter-spike intervals for 200 sampled neurons per layer, accumulated over the validation partition.

5.4 What the explanations show

Figure 5.5 shows the SAM sequence of a single Blip at the reference layer. The map is zero at $t = 0$ by definition and builds up over the first steps as spike histories accumulate; from the third step onward the attended region is the vertical stroke of the glitch.

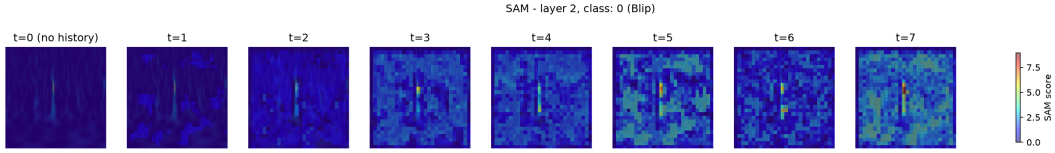


Figure 5.5: SAM at layer 2 for one Blip, one panel per time step, on a shared colour scale.

The class statistics (Figure 5.6) show four recognizably different attention patterns, each matching the morphology of its class. For Blip the mean map concentrates on the narrow vertical stroke and so does the variability, which is the stroke shifting slightly between instances. For Extremely_Loud the attended region is large and saturated, covering the lower and central part of the spectrogram. For Scattered_Light the attention lies on a horizontal band at low frequency. Violin_Mode produces the weakest and most diffuse mean map, with variability concentrated in a few localized spots along the line structure; it is also the class with the lowest recall, and the one whose explanations prove most sensitive in every test below.

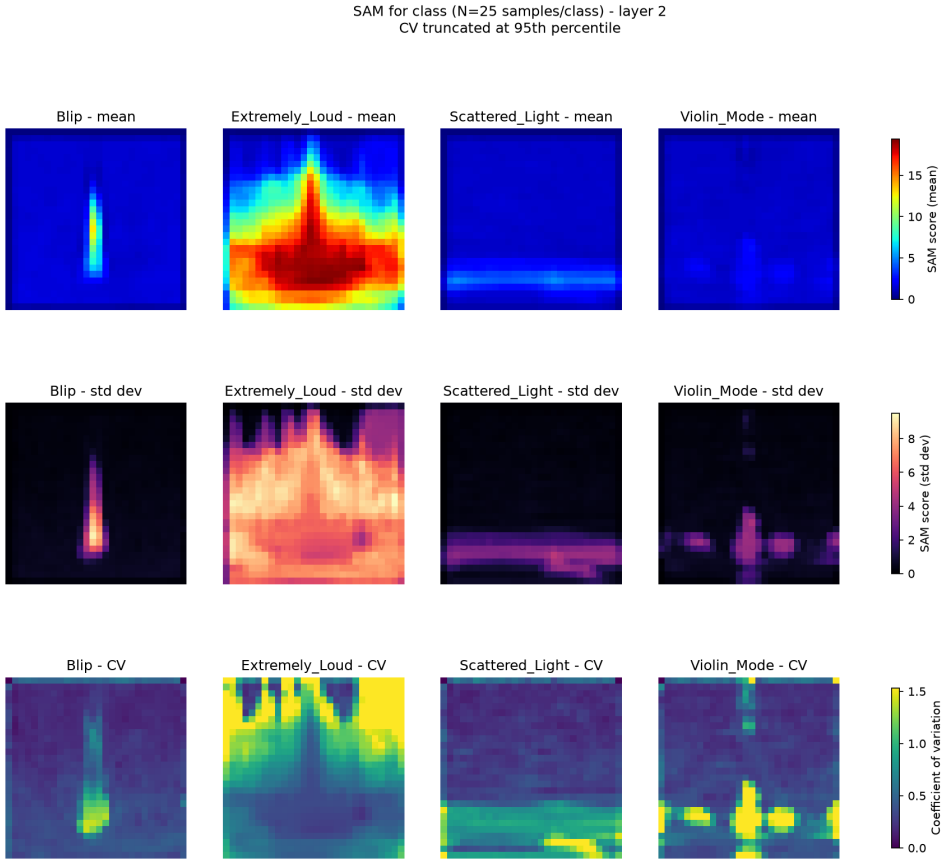
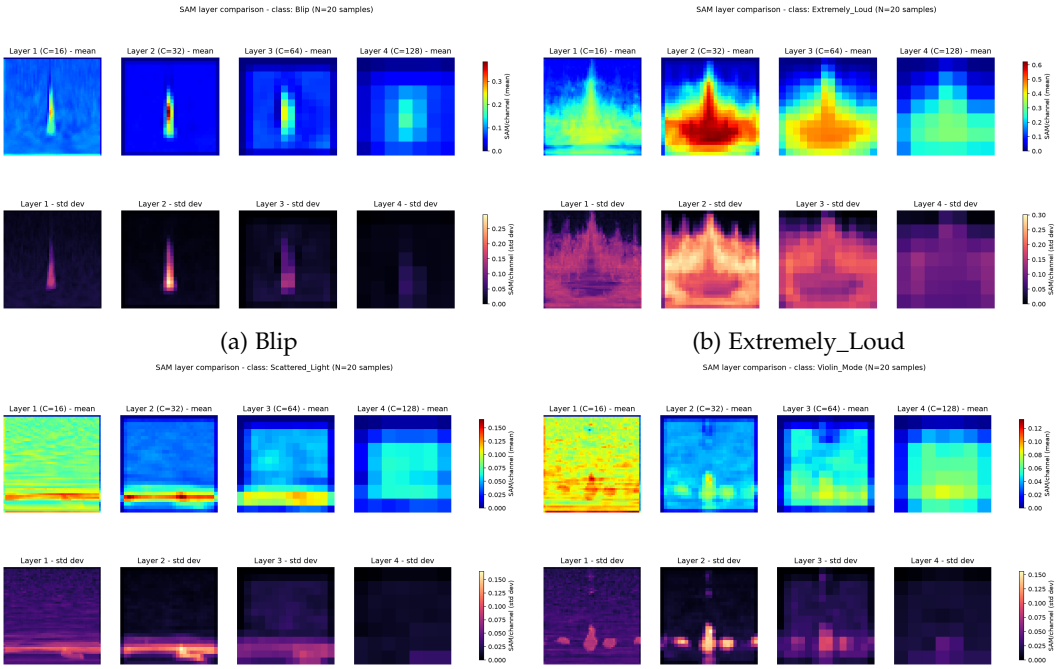


Figure 5.6: SAM statistics per class at layer 2, 25 validation images per class: mean (top), standard deviation (middle) and coefficient of variation (bottom, truncated at the 95th percentile). Mean and standard deviation share a colour scale across classes.

Across depth (Figure 5.7) the progression is the expected one, from fine localization in the early layers to coarse localization in the deep ones, with maps of side 56, 28, 14 and 7. At layer 4 a 7×7 map leaves only 49 positions, too few for a top-20 % overlap or a structural comparison to be meaningful, while layer 2 still resolves the morphology of every class.

TO DO. Confirm (or replace) this as the reason for choosing layer 2 as the reference layer; the progress report records the choice but not its justification.



(a) Blip

(b) Extremely_Loud

(c) Scattered_Light

(d) Violin_Mode

Figure 5.7: SAM mean and standard deviation across the four hidden layers, 20 images per class, normalized by the channel count of each layer.

5.5 Fixing the explainer

5.5.1 Choice of the decay rate

SAM has one free parameter, and an explanation that changed qualitatively with an arbitrary choice of γ would not be trustworthy.

γ is the decay of a temporal kernel, so it must be evaluated on the temporal axis. Comparing maps averaged over time removes precisely the axis on which γ acts, and a high spatial correlation is then close to guaranteed regardless of γ. Three quantities are therefore measured: the spatial Pearson correlation between time-aggregated maps, retained for continuity with the literature; the *temporal centre of mass* of the SAM energy; and the *per-timestep correlation* against a reference γ.

Before a null result could be trusted, the instrument was checked analytically: on a fully firing spike train with $T = 8$ the centre of mass moves from 4.86 at $\gamma = 0.1$ to 4.02 at $\gamma = 3.0$, so the metric does respond. A second check comes free from the definition: at $t = 1$ the score is $e^{-\gamma} S_0$, so changing γ multiplies the map by a constant, and Pearson correlation is affine-invariant. The measured per-timestep correlation at $t = 1$ is exactly 1.000 for every γ, as it must be.

Class	Spatial PCC, γ 0.1 vs 3.0	CoM at 0.1	at 3.0	Shift
Extremely_Loud	0.9993	4.973	4.144	0.829
Blip	0.9890	5.192	4.339	0.853
Scattered_Light	0.9758	5.271	4.352	0.919
Violin_Mode	0.9598	5.260	4.327	0.933

Table 5.4: γ sensitivity at layer 2, mean over fifteen validation images per class.

Two conclusions follow, and neither is the one the naive analysis would reach.

The temporal shift is largely kernel arithmetic rather than explanation structure. The measured shift of 0.83–0.93 steps matches what a *fully firing* spike train produces (0.84), whereas the layer-2 firing rate is approximately 0.19 and simulated random trains at that density shift by only 0.26 steps. SAM is therefore dominated by a minority of near-continuously firing neurons, for which the centre of mass follows the exponential kernel. The centre of mass against γ is consequently *not* a useful criterion for selecting γ — though it remains valid as a divergence metric, where models are compared at fixed γ and the trivial component cancels, and it now carries a reference scale.

Sensitivity is strongly class-dependent. In terms of $1 - \mathrm{PCC}$, Violin_Mode is 57 times more sensitive than Extremely_Loud. The ordering Violin_Mode > Scattered_Light > Blip > Extremely_Loud recurs identically under the two unrelated interventions of Section 5.6.

Within $\gamma \in [0.3, 0.9]$ all per-timestep correlations remain above 0.98 even in the worst class; only $\gamma = 3.0$ diverges appreciably. $\gamma = 0.5$ **is adopted**, the value of the original formulation [22], now supported by evidence on the axis γ actually governs (Figure 5.8).

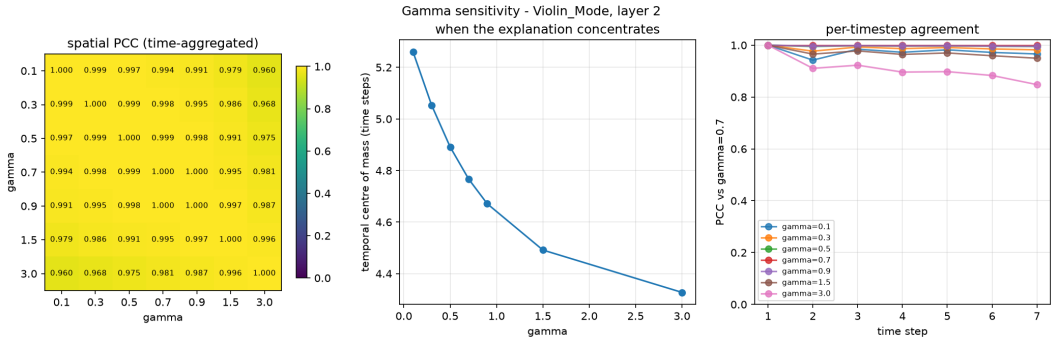


Figure 5.8: γ sensitivity for Violin_Mode, the most sensitive class, at layer 2. The equivalent figures for the other three classes are in Appendix A.

5.5.2 Faithfulness

Robustness to γ is necessary but not sufficient: a stable explanation can be stably wrong. The deletion metric [32] removes pixels in decreasing order of SAM score and records the confidence assigned to the class predicted on the unmodified image, comparing against removal in random order.

Three implementation details govern interpretation. Deleted pixels are set to zero *in normalized space*, corresponding to the dataset mean rather than to black, a neutral substrate that avoids injecting an artificial edge. The same map is used for both orderings. And the ordering is computed once per image and sliced progressively, so the masked sets are nested: what is removed at 50 % stays removed at 55 %. Regenerating a random permutation at each fraction would break nesting for the baseline while leaving it intact for SAM, making the two curves incomparable.

Class	AUC-SAM	AUC-random	Δ	Relative	Reading
Blip	0.3119	0.4795	−0.1676	−35.0 %	faithful
Extremely_Loud	0.6121	0.8892	−0.2771	−31.2 %	faithful
Scattered_Light	0.3835	0.3650	+0.0185	+5.1 %	not separable
Violin_Mode	0.4528	0.4540	−0.0012	−0.3 %	indistinguishable

Table 5.5: Deletion metric by class, 15 validation images each. A faithful explanation gives AUC-SAM below AUC-random, so a negative Δ is expected.

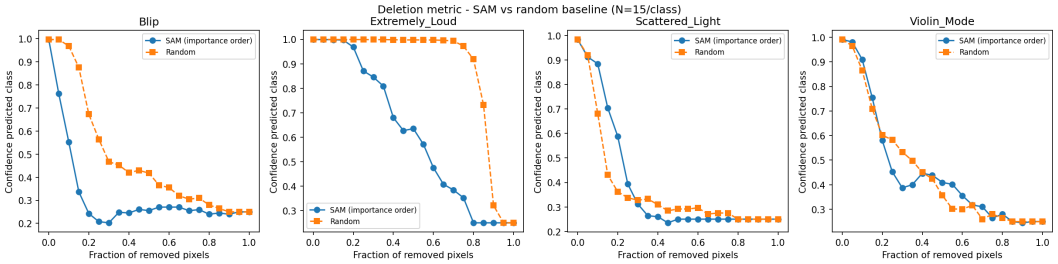


Figure 5.9: Deletion curves, SAM order against random order, 15 validation images per class.

The metric separates SAM from the baseline on two classes and not on the other two, and the split is systematic. Blip and Extremely_Loud show reductions of 35 and 31 % in area. Violin_Mode is within 0.3 % of the baseline. Scattered_Light is nominally inverted, but at +5.1 % the magnitude is one sixth of the two positive effects, so it is better described as an absence of separation with a slight drift in the wrong direction than as a genuine reversal.

The pattern follows the spatial geometry of the classes rather than any property of SAM. Blip is a compact vertical line and Extremely_Loud a dominant saturated structure, so removing the high-scoring region removes the defining evidence. Scattered_Light is an extended horizontal band and Violin_Mode a set of thin distributed lines: random deletion fragments the pattern across the whole image, whereas SAM-guided deletion concentrates removal in one region and leaves the rest — which still carries sufficient evidence — intact. This is a known limitation of deletion-based tests, which penalize explanations of spatially extended features, and it is reported rather than suppressed.

Two controls confirm the measurement itself. All curves converge to approximately 0.25 at full deletion, which is chance for four classes. And the Extremely_Loud random baseline remaining at 0.8892, the highest of the four, is independent confirmation of that class's behaviour below: its evidence is distributed across the whole image, which is why no localized removal registers.

5.6 Calibrating the divergence metrics

The metrics used to compare two explanations return numbers between zero and one whatever is fed to them. Three questions must be answered before any quantization result can be interpreted: what they report when nothing has changed, when something negligible has changed, and when everything has changed.

All measurements use fifteen validation images per class at layer 2 with $\gamma = 0.5$, on the three checkpoints of Table 5.3.

5.6.1 Verification of the instrument

Comparing a checkpoint against itself returns PCC $= 1$, IoU $= 1$ and a centre-of-mass difference of 0 to within 10^{-9}, confirming that the metrics and the image pairing are correct. Two separate calls to the sampling routine return bit-identical image tensors, confirming that every model sees the same data. A second forward pass of the same model reproduces the maps exactly, confirming that the network is deterministic in evaluation mode and that the cross-model differences below are not run-to-run noise.

Class	PCC mean	worst	IoU mean	worst	CoM diff, max
Extremely_Loud	0.917	0.869	0.520	0.460	0.076
Violin_Mode	0.844	0.816	0.597	0.543	0.115
Blip	0.612	0.336	0.593	0.503	0.083
Scattered_Light	0.477	**0.153**	0.376	0.234	0.104

Table 5.6: Divergence between full-precision models differing only in seed, over the three seed pairs. Chance-level IoU at $k = 0.20$ is 0.111.

5.6.2 Two independent solutions

The spatial metrics collapse. Two models agreeing to within one validation sample in accuracy produce maps correlating at 0.153 on Scattered_Light, with a top-20 % overlap of 0.234 against a chance level of 0.111. For comparison, a thirtyfold change in γ reduces the correlation only to 0.960: initialization variance is 21 times larger than that effect.

The per-image distribution rules out an outlier explanation. On Blip the fifteen per-image correlations run from 0.147 to 0.768 with a median of 0.282, and twelve of the fifteen fall below 0.35. The disagreement is uniform across samples.

The mechanism is not a defect of the metric. Networks trained from different initializations learn functionally equivalent but **permuted** representations, so a pixel-wise comparison between them compares arbitrary assignments of channels to positions rather than information content. This is why the result is a bound on cross-model spatial claims rather than the threshold for the quantization grid, where the quantized model inherits its parent's weights and no permutation exists.

The temporal centre of mass survives. The maximum difference across seed pairs is 0.115 time steps, against the ≈ 0.90 produced by a thirtyfold change in γ — a factor of eight. Being an aggregate over the whole map, it is insensitive to which neuron occupies which position, which is precisely what defeats the pixel-wise metrics.

5.6.3 A negligible perturbation

The grid compares a quantized model with its own parent, so the operative threshold is what a perturbation far smaller than quantization already produces. Gaussian noise was added to the weight tensors of one checkpoint — biases and LIF buffers excluded, since perturbing β would confound membrane dynamics with weight quantization — at magnitudes expressed as fractions of the measured 8-bit step of those weights (0.00168, from a maximum absolute weight of 0.214).

This is a graded form of the model randomization test of Adebayo et al. [4]: where they randomize weights to verify that a metric responds at all, this perturbs towards zero to find where it stops responding.

Noise / 8-bit step	Extremely_Loud	Blip	Scattered_Light	Violin_Mode
0.01	1.0000	0.9997	0.9996	0.9991
0.1	0.9999	0.9981	0.9977	0.9930
0.5	0.9998	0.9949	0.9938	0.9824
1.0	0.9997	0.9921	0.9901	0.9749

Table 5.7: Spatial PCC against the unperturbed model. The 4-bit and 2-bit quantization steps are 17 and 85 times larger than the largest perturbation shown.

The spatial metrics are usable after all, for same-parent comparisons. PCC remains above 0.975 under a perturbation as large as the entire 8-bit step. The collapse of Table 5.6 is a consequence of comparing different initializations, not a fragility of the metric; the two results answer different questions.

The metrics differ sharply in dynamic range. Measuring the distance from perfect agreement, PCC starts 0.025 away from 1 and can reach 0.847: 34 times its own floor. IoU starts 0.296 away and reaches 0.766, a factor of only 2.6, and is already at 0.704 under a perturbation 85 times smaller than the 2-bit step. IoU should therefore be read as a sensitive early indicator that a change has occurred, and PCC as the graded measure of how large it is. The temporal centre of mass sits between them, with a factor of 3.9 between its numerical floor (0.029 steps) and the seed level (0.115).

Extremely_Loud does not respond. Its PCC is 0.9997 under the largest perturbation and its worst per-timestep correlation is 0.996. Combined with its behaviour under γ, this class cannot register an explanation change by any measure tested here. A flat grid result for Extremely_Loud will not distinguish absence of effect from inability to detect one, and this is stated in advance.

5.6.4 Fine-tuning alone

Every quantized model in the grid is its parent with bits removed *and* then fine-tuned. Part of any divergence measured against that parent belongs to the fine-tuning. The three levels above cannot separate the two contributions: the numerical floor is a perturbed copy that was never trained, and the independent-solutions level is two networks grown from different initializations. What is missing is the model that went through the whole procedure with nothing removed.

The control arm supplies it, but not through the checkpoint the grid uses. That checkpoint is the parent (Section 4.1) and returns 1.000 on every class and every seed, which measures nothing. The weights after the last epoch are a different model: the parameter vector moves by an L_2 distance of 0.395, 0.736 and 0.399 over the three seeds, nine tenths of it in the last convolution and the linear readout.

Seed	Epochs	$\|\Delta w\|_2$	Accuracy	IoU	PCC	Worst class IoU
1234	6	0.395	0.9819	0.711	0.985	0.671
2345	14	0.736	0.9791	0.792	0.989	0.779
3456	6	0.399	0.9805	0.748	0.987	0.705
Mean			0.9805	**0.750**	0.987	0.671

Table 5.8: The control arm after its last fine-tuning epoch, against its own full-precision parent, twenty validation images per class at layer 2, $\gamma = 0.5$. No bits are removed in any of these runs.

Six to fourteen epochs with the learning rate scaled by 0.1, and no bits removed, move the top-20 % overlap to 0.750, with a seed range of 0.711 to 0.792. Validation accuracy falls by 0.3 points over the same runs, so the movement is not a model that degraded; it is a model that kept training.

Two details are worth recording. The seed whose weights moved furthest, 2345 at 0.736, is the seed whose explanation moved *least*, at 0.792: over three runs this is an observation rather than a result, but distance in parameter space does not predict distance in explanation space, which is the premise the grid rests on. And the Pearson correlation of this anchor is 0.987 against an IoU of 0.750 — the metric disagreement that recurs throughout the grid appears here too, where there is no quantization to attribute it to.

5.6.5 What the calibration establishes

The *numerical floor* marks measurement noise: a divergence below what a negligible weight perturbation already produces cannot be read at all. The operative threshold for the grid is the one above it. *Fine-tuning alone* is what the procedure costs when nothing is quantized, so a configuration that diverges by less than 0.750 has not been moved by the removed

Level	What it measures	PCC	IoU	CoM
Numerical floor	Weight noise at $1\times$ the 8-bit step	0.975	0.704	0.029
Fine-tuning alone	The procedure with no bits removed	0.987	0.750	—
γ reference	A thirtyfold change in the kernel	0.960	—	0.90
Independent solutions	Two FP32 models, different seeds	0.153	0.234	0.115

Table 5.9: The four calibration levels, worst class in each case, at 112 px, layer 2, $\gamma = 0.5$, on validation. The first, third and fourth use fifteen images per class, the fine-tuning level twenty, matching the grid it is read against.

bits in any way this experiment can establish — it has been moved by the training that followed them. The γ *reference* gives an intuition: if quantization produced a PCC of 0.90, it would be doing more than a thirtyfold change of the explainer itself. The *independent-solutions* level is a pessimistic upper bound and a result in its own right.

The class sensitivity ordering — Violin_Mode > Scattered_Light > Blip > Extremely_Loud — recurs identically under three unrelated interventions: changing the kernel, changing the initialization, and perturbing the weights. Explanation sensitivity is therefore a property of the class morphology, and every grid result must be reported per class, since a pooled figure would average a class that responds strongly with one that does not respond at all.

5.7 Preparing the membrane arm

5.7.1 The range of the membrane potential

The membrane potential of every LIF layer was recorded at every time step over the full validation partition (1 435 images, eight steps) on the reference checkpoint. Table 5.10 summarizes the distributions.

Layer	min	$p_{0.1}$	median	$p_{99.9}$	max	std	$> U_{\text{thr}}$
1	−12.75	−9.43	−0.08	6.40	8.76	1.98	14.9 %
2	−15.73	−11.10	0.25	5.63	11.64	1.42	15.1 %
3	−47.03	−36.82	0.00	11.27	20.23	4.22	15.3 %
4	−26.04	−17.01	−0.62	6.34	14.67	2.59	11.5 %

Table 5.10: Membrane potential of the four hidden layers over the validation partition, reference checkpoint (seed 1234). The last column is the fraction of values above the firing threshold $U_{\text{thr}} = 1$, which equals the mean firing rate of the layer.

The distributions are wide and strongly skewed towards negative values: in layer 3 the potential reaches −47, almost fifty thresholds below rest. A grid covering the full observed range would be dominated by values far from the threshold, which is the only point at which a potential affects a spike decision. At 4 bits, a symmetric grid over $[-47, +47]$ would have a step of more than six thresholds, and the network could not be represented at all.

5.7.2 Clipping the potential costs nothing

Before any quantization, the potential of the four hidden layers was *clipped* at every time step to a candidate range, on the full-precision network, and the effect on accuracy and dynamics was measured. Clipping isolates the choice of range from the choice of bit-

width: if a range already changes the network in full precision, no grid built on it can be faithful.

Clipping range	Val. accuracy	Δ samples	Layer-4 CoM
none	0.98328	—	3.9469
per layer, $[p_{0.1}, p_{99.9}]$	0.98328	0	3.9469
per layer, $[p_1, p_{99}]$	0.98328	0	3.9469
fixed $[-8, +4]$	0.98328	0	3.9469
fixed $[-4, +4]$	0.98328	0	3.9469
fixed $[-2, +2]$	0.98258	-1	3.9415
unsigned $[0, +4]$	0.98258	-1	3.9483
unsigned $[0, +2]$	**0.98258**	$\mathbf{-1}$	**3.9431**
control $[-0.5, +0.5]$	0.84808	-194	4.0684

Table 5.11: Effect of clipping the hidden membrane potentials on the full-precision reference (seed 1234), on validation. The tolerance is the spread of the three-seed reference, four samples out of 1 435. The last column is the temporal centre of mass of the layer-4 spike counts.

Three observations follow from Table 5.11.

Negative potentials carry no information the network uses. Clipping to $[0, 2]$ removes every negative value — in layer 3 a range of 47 thresholds — and costs one validation sample out of 1 435, a quarter of the spread between seeds. The per-layer dead fractions are unchanged to the first decimal (16.1 / 12.5 / 11.3 / 22.5 % against 16.1 / 12.6 / 11.3 / 22.8 %), and the largest shift in the temporal centre of mass of any layer's spike counts is 0.004 steps, well below the 0.029-step floor measured for the SAM centre of mass in Section 5.6 (the two quantities are related but not identical). Below rest, how far below no longer matters: a neuron at -20 and a neuron at 0 both need more than one threshold of input to fire.

The upper bound can be tight. Potentials above twice the threshold are rare and can be clipped at no measurable cost. Since the clipping acts after the spike decision, it only alters the carried state of neurons that have just fired well above threshold; the spike they emitted at that step is unaffected.

The clipping is not inert. The control range $[-0.5, +0.5]$, which keeps the potential permanently below threshold after each step, drops accuracy to 0.848 and raises the layer-3 firing rate from 0.153 to 0.227. The hook therefore acts on the state the network actually carries, and the null results above are genuine.

The unsigned range $[0, 2]$ is adopted: it is the tightest range tested that stays within tolerance, it centres the threshold in the grid, and it spends no levels on negative values, which saves one bit at every bit-width compared with the symmetric $[-2, 2]$ that performs identically.

5.8 Energy and memory of the reference

Table 5.12 applies the accounting of Section 4.8 to the full-precision reference, using the per-hidden-layer firing rates averaged over the three seeds (0.258/0.186/0.185/0.126).

The spiking network costs 62.8 µJ per inference, 2.5 times less than a convolutional network of identical shape. The saving depends entirely on the treatment of the first layer: recomputing its constant current at every step would cost 183.9 µJ, more than the convolutional equivalent, and the two readings differ by a factor of 2.9. Even computed once, the first convolution is 28 % of the budget, because it is the only layer that multiplies.

The initialization regime of Section 5.2 reaches the energy as well. Computed seed by seed, the estimate is 50.0, 78.1 and 60.1 µJ for seeds 1234, 2345 and 3456, the difference

Layer	MAC (dense)	Operations	Energy (µJ)
conv1 (analog input, once)	3.76×10^6	3.76×10^6 MAC	17.31
conv2	10.04×10^6	20.72×10^6 SOP	18.65
conv3	10.04×10^6	14.93×10^6 SOP	13.44
conv4	10.04×10^6	14.83×10^6 SOP	13.34
linear	0.03×10^6	0.03×10^6 SOP	0.02
SNN, conv1 once			**62.8**
SNN, conv1 at every step			183.9
Equivalent CNN (all MAC)	33.89×10^6		155.9

Table 5.12: Estimated energy per inference of the FP32 reference at 112 px, T = 8, with 45 nm costs of 4.6 pJ per MAC and 0.9 pJ per accumulate.

coming almost entirely from the first two layers, where the firing regime varies most. Three models indistinguishable in accuracy therefore differ by more than half in estimated energy, and every energy comparison in the grid is made at matched seed.

In memory, the 295 088 weights occupy 1 152.7 KiB in 32-bit float (the 244 biases, kept in full precision, are negligible), and the membrane state of the 94 080 hidden neurons another 367.5 KiB. At 8, 4 and 2 bits these become 288, 144 and 72 KiB for the weights and 92, 46 and 23 KiB for the state.

5.9 The quantization grid

All thirty fine-tuning runs completed without failure. Unless stated otherwise, every figure below is a mean over the three seeds, each quantized model being compared with the full-precision parent of its own seed, on validation. Figure 5.10 summarizes the grid; the sections that follow take it apart.

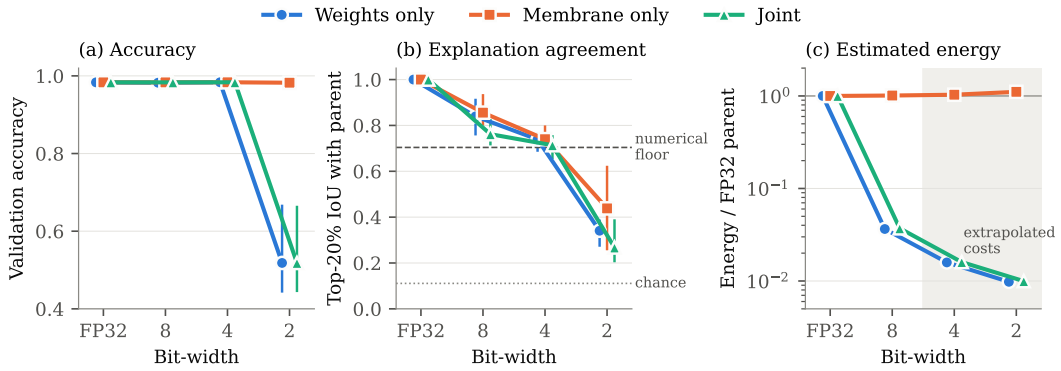


Figure 5.10: The grid on a common bit-width axis. (a) Validation accuracy after fine-tuning. (b) Top-20 % IoU between the SAM maps of each quantized model and those of its parent, mean over the four classes; the dashed line is the numerical floor of the calibration, the dotted line chance. (c) Estimated energy per inference relative to the parent, logarithmic scale; the shaded region uses extrapolated per-operation costs. Points are means over three seeds, bars span the seeds. Markers are offset horizontally for legibility.

5.9.1 Gates and control

The quantizer passed both gates. Attaching and removing it restores the original weights exactly, and a quantized checkpoint reloads with zero drift. With no fine-tuning, the joint 8-bit model of seed 1234 reaches 0.98328 validation accuracy, identical to its parent; the 8-bit weight arm loses three samples out of 1 435, inside the four-sample tolerance.

The control arm, which runs the full fine-tuning procedure with no quantizer, returns a spatial correlation of exactly 1, an IoU of exactly 1 and a centre-of-mass shift of exactly 0 with its parent, in every class and every seed. Fine-tuning at the reduced learning rate never improved on the parent's validation accuracy, so the selected control checkpoint is the parent itself: the pipeline — reload, fine-tune, select, re-evaluate — introduces nothing when nothing is quantized.

The guard of Section 4.6 holds as well: the 8-bit weight arm keeps a spatial correlation of at least 0.982 with its parent in every class of every seed.

5.9.2 Accuracy

Arm	Bits	PTQ	After fine-tuning	Δ samples	Epochs
Weights	8	0.9819	0.9823 ± 0.0008	-1.3	9.7
Membrane	8	0.9814	0.9833 ± 0.0000	0.0	8.3
Joint	8	0.9819	0.9833 ± 0.0007	0.0	8.0
Weights	4	0.9547	0.9835 ± 0.0016	$+0.3$	11.7
Membrane	4	0.9807	0.9835 ± 0.0011	$+0.3$	10.3
Joint	4	0.9508	0.9835 ± 0.0015	$+0.3$	14.0
Weights	2	0.3591	0.5182 ± 0.1300	-667	18.3
Membrane	2	0.9517	$\mathbf{0.9823 \pm 0.0004}$	-1.3	13.0
Joint	2	0.3652	0.5182 ± 0.1275	-667	19.0

Table 5.13: Validation accuracy across the grid: post-training quantization (PTQ, no fine-tuning) and after fine-tuning, mean $\pm$ standard deviation over three seeds. Δ is the difference from the FP32 reference (0.9833 ± 0.0014) in validation samples out of 1 435.

At 8 and 4 bits accuracy is indistinguishable from full precision in every arm (Table 5.13); every difference is below one validation sample on average. Fine-tuning matters at 4 bits: post-training quantization of the weights costs about 40 samples, and fine-tuning recovers all of them.

At 2 bits the arms separate completely. The membrane arm keeps parity, 0.9823, within three samples of its parent in every seed. The weight and joint arms collapse to 0.518, and the collapse is not uniform across classes: no 2-bit weight model ever predicts Scattered_Light, and two of the three never predict Violin_Mode either, assigning every image to Blip or Extremely_Loud. Fine-tuning recovers 16 points over post-training quantization but cannot recover a ternary network. A 2-bit membrane, with a single level between rest and threshold, is enough to classify; ternary weights are not.

The asymmetry inverts the prediction derived from Smith et al. [38] in Section 4.4, that the membrane would need at least four bits while weights would survive two. In this network and with this grid, it is the other way round.

5.9.3 Explanation divergence

At 8 and 4 bits the explanations stay at the floor. Spatial correlation remains above 0.977 on average in every arm, and the IoU between 0.71 and 0.86, around the numerical floor of 0.704: at 4 bits four of the twelve class-and-seed cells of each arm fall marginally below it, the lowest at 0.595, all far above the independent-solutions level of 0.234 (Tables 5.14

Arm	Bits	Blip	Extremely_Loud	Scattered_Light	Violin_Mode	Mean	Worst cell
Weights	8	0.821	0.854	0.841	0.841	0.839	0.716
Membrane	8	0.845	0.876	0.843	0.858	0.856	0.766
Joint	8	0.753	0.779	0.749	0.766	0.762	0.674
Weights	4	0.719	0.787	0.717	0.733	0.739	0.649
Membrane	4	0.726	0.773	0.719	0.743	0.740	0.628
Joint	4	0.696	0.752	0.691	0.715	0.714	0.595
Weights	2	0.269	0.335	0.415	0.344	0.341	0.127
Membrane	2	**0.300**	**0.559**	**0.490**	**0.404**	**0.438**	**0.102**
Joint	2	0.217	0.339	0.344	0.171	0.268	0.057

Table 5.14: Top-20 % IoU between the SAM maps of each quantized model and those of its parent, layer 2, $\gamma = 0.5$, twenty validation images per class, mean over three seeds. "Worst cell" is the lowest class-and-seed value. Numerical floor 0.704, chance 0.111.

Arm	Bits	PCC mean	PCC worst	CoM $\lvert\Delta\rvert$ mean	CoM $\lvert\Delta\rvert$ worst	CoM signed
Weights	8	0.995	0.982	0.007	0.015	-0.005
Membrane	8	0.995	0.989	0.010	0.025	-0.009
Joint	8	0.989	0.973	0.022	0.049	-0.022
Weights	4	0.982	0.960	0.049	0.094	-0.048
Membrane	4	0.985	0.963	0.044	0.064	$+0.043$
Joint	4	0.977	0.949	0.021	0.043	$+0.018$
Weights	2	0.827	0.748	0.211	0.369	-0.211
Membrane	2	0.923	0.799	0.068	0.137	-0.053
Joint	2	0.780	0.580	0.235	0.367	-0.235

Table 5.15: Spatial Pearson correlation and temporal centre-of-mass shift (in time steps) against the parent, same protocol as Table 5.14. A negative signed shift means that the spike energy of the explanation moves earlier. Numerical floors: PCC 0.975, CoM 0.029; independent-solutions level: PCC 0.153, CoM 0.115.

and 5.15). The temporal centre of mass is the only metric that moves at 4 bits, by 0.044 to 0.049 steps in the two single-target arms, above its floor of 0.029 and below the 0.115 that separates independent solutions. The direction is informative: quantizing the weights moves the spike energy of the explanation *earlier*, quantizing the membrane moves it *later*, and in the joint arm the two shifts partly cancel (+0.018).

At 2 bits the weight and joint arms explain a different classifier. Their IoU falls to 0.34 and 0.27 and their centre of mass shifts by more than the independent-solutions level. Since those models no longer classify, the divergence of their explanations is a consequence of the collapse rather than a measurement of explanation fidelity at constant accuracy, and it is not read as such.

The membrane arm at 2 bits is the result the grid was built to find. Accuracy is at parity, yet the IoU with the parent's explanation is 0.438, 38 % below the numerical floor. All twelve class-and-seed cells are below the floor, and two are below chance (the lowest 0.102): for those, the 20 % of the image that SAM highlights in the quantized model overlaps the parent's no more than two random selections would.

The comparison that carries the thesis is between two models with the same accuracy. The 4-bit weight model and the 2-bit membrane model both reach 0.983 (0.9835 and 0.9823). The first explains its decisions with essentially the same pixels as its parent, IoU 0.739; the second shares less than half of them, IoU 0.438. Selected on accuracy, the two are interchangeable; their explanations are not.

The spatial correlation alone would have given the wrong answer. The 2-bit membrane model scores a PCC of 0.923. Against the values that the literature presents as preserved explanations, around 0.85, that figure would be reported as a surviving explanation; it is below the numerical floor, but by an amount that understates what changed. Pearson correlation is dominated by the broad, low-intensity structure that every map of a class shares, while the top-k overlap measures the region a reader is actually shown. This is the disagreement between metrics that Bylinskii et al. [8] warn of, and it is why both are reported.

The effect depends on the seed. The 2-bit membrane IoU is 0.255, 0.624 and 0.435 for the three seeds, a spread larger than any effect measured within a seed below. The class ordering is also not the one of the calibration: Blip, not Violin_Mode, loses the most (0.300), and Extremely_Loud, which registered no perturbation in the calibration, now moves (0.559).

5.9.4 The hypothesis

The hypothesis predicted an ordering: the membrane arm displaces the temporal centre of mass at a higher bit-width than the one at which the weight arm lowers the spatial correlation. The grid does not show that ordering. Both effects first appear at the same bit-width, 4 bits, where the weight arm lowers the PCC below the per-class floor on every class and both single-target arms shift the centre of mass by the same amount. At 2 bits, where the arms do separate, the visible effect of membrane quantization on the explanation is *spatial* — which pixels are highlighted — while its centre-of-mass shift (0.068) stays below the independent-solutions level. **As formulated in terms of SAM metrics, the hypothesis is not supported.**

Its mechanism, however, is visible elsewhere. The per-model validation pass measures the regularity of firing, and there the membrane arm at 2 bits does something no other configuration does.

In every model with a 2-bit membrane, the inter-spike intervals of layer 1 become perfectly regular (Table 5.16): the CV is exactly zero in all six measurements, while it is unchanged with 2-bit weights and with a 4-bit membrane. The fraction of neurons for which the CV is defined is comparable in every configuration, so the effect is not a change in which neurons are measured. The mechanism follows from Table 4.3: with a single level

Configuration	Seed 1234	Seed 2345	Seed 3456	Neurons with defined CV
FP32 parent	0.132	0.241	0.051	2.7 / 24.1 / 10.4 %
Membrane, 4 bit	0.134	0.236	0.051	2.5 / 23.4 / 10.3 %
Weights, 2 bit	0.140	0.104	0.208	4.0 / 25.3 / 10.0 %
Membrane, 2 bit	**0.000**	**0.000**	**0.000**	3.2 / 28.5 / 10.7 %
Joint, 2 bit	**0.000**	**0.000**	**0.000**	5.1 / 27.2 / 10.6 %

Table 5.16: Mean coefficient of variation of the inter-spike intervals in layer 1, per seed, and the fraction of neurons for which it is defined.

between rest and threshold, the state of a neuron between two spikes can take only one intermediate value, so under the constant drive of direct encoding the number of steps from reset to threshold cannot vary, and firing becomes periodic. Membrane quantization does act on the temporal dynamics first, as the hypothesis argued; but the change is in the *regularity* of the spike trains, and the temporal centre of mass, which measures where the spike mass falls in time and not how regularly it arrives, cannot see it.

The other diagnostics behave accordingly. The 2-bit membrane raises the firing rate of every layer (from 0.258/0.186/0.185/0.126 to 0.273/0.218/0.229/0.151) with little change in the dead fraction, whereas 2-bit weights double the dead fraction of layer 4 (from 23.6 to 43.1 %) and saturate up to 6.8 % of layer 1.

5.9.5 Faithfulness

The deletion metric was repeated on the 2-bit membrane models, the only configuration with intact accuracy and a changed explanation, with the baseline recomputed on the parent of each seed (Table 5.17).

Seed	Class	Parent baseline	Own map	Parent's map
1234	Blip	0.153	0.166	0.117
1234	Extremely_Loud	0.255	0.248	0.236
2345	Blip	0.087	0.083	0.099
2345	Extremely_Loud	0.222	0.136	0.149
2345	Scattered_Light	0.056	0.074	0.083
2345	Violin_Mode	0.106	0.187	0.193
3456	Blip	0.111	**0.033**	0.037
3456	Extremely_Loud	0.207	0.186	0.180
3456	Scattered_Light	0.112	0.094	0.056

Table 5.17: Deletion gap (AUC of random deletion minus AUC of SAM-guided deletion; larger means more faithful) for the 2-bit membrane models, fifteen validation images per class. Only the nine class-and-seed cells whose parent baseline is at least 0.05 are shown; the other three have no signal to degrade.

The explanation is changed, not broken. In every interpretable cell the quantized model's own map still separates from random deletion; the median gap is 0.95 of the parent's, and only one cell comes close to losing it — Blip on seed 3456, from 0.111 to 0.033. Explaining the quantized model with its parent's maps works as well as with its own: the mean difference between the two gaps is +0.0065 over the nine cells, four positive and five negative, never more than 0.049, although the two maps share only 44 % of their highlighted pixels.

The two measurements are therefore not in contradiction; they answer different questions. The deletion metric rewards any map that points at evidence the model uses, and when the evidence is redundant — a Blip stroke covers many positions, any large part of which suffices — several disjoint sets of pixels satisfy it. The IoU says which of those sets

a reader is shown. A commissioner looking at the quantized model's map would be sent to a region that does carry evidence, but not to the one the original model would have shown.

Which classes SAM can explain at all depends on the seed. Only nine of the twelve class-and-seed cells have a baseline signal: Violin_Mode is explainable in one parent of three, Scattered_Light in two of three, and even Blip's baseline gap ranges from 0.087 to 0.153 across networks of equal accuracy. The set of classes on which faithfulness can be assessed is a property of the individual network, not of the class.

5.9.6 Stratification by event strength

	Membrane, 2 bit			Membrane, 4 bit		
Class	1234	2345	3456	1234	2345	3456
Blip	+0.121	+0.038	+0.164	+0.060	+0.012	+0.039
Scattered_Light	+0.443	−0.021	+0.138	+0.242	+0.017	+0.109
Violin_Mode	+0.134	−0.038	−0.124	+0.093	+0.022	+0.028
Mean of the three	+0.233	−0.007	+0.059	+0.132	+0.017	+0.059
Extremely_Loud	−0.469	−0.202	−0.354	−0.234	−0.101	−0.142

Table 5.18: IoU gradient with event strength: IoU in the high-SNR tercile minus IoU in the low-SNR tercile, terciles computed within each class, fifteen images per cell. A positive value means that weak events lose their explanation more.

The data do not support the hypothesis that degradation concentrates on weak glitches. The mean gradient over Blip, Scattered_Light and Violin_Mode is +0.233, −0.007 and +0.059 in the three seeds at 2 bits (Table 5.18): positive on average, but the spread between seeds is as large as the effect, and the between-seed range of the 2-bit IoU itself (0.255 to 0.624) exceeds the largest within-seed gradient.

One effect is consistent, and of opposite sign. On Extremely_Loud the gradient is negative in all six measurements: the *strongest* events lose their explanation most. These are the events that saturate the spectrogram, so that most of the image carries evidence and the top-20 % region is least constrained; a small change of the network is enough to move it.

5.9.7 Energy and memory

The membrane arm does not save energy; it costs more. Its weights stay in full precision, so the cost of each operation never falls, while the quantized membrane raises the firing rates and with them the number of operations: +1.1 % at 8 bits, +3.1 % at 4, +10.8 % at 2 (Table 5.19). What it does save is state memory, 344 KiB at 2 bits, almost a quarter of the total footprint.

Quantization raises the operation count in every arm, mildly at 8 and 4 bits (1–8 %) and sharply at 2 bits, where it grows by half with quantized weights: the collapsed networks fire more, not less. In the weight and joint arms the saving still comes entirely from the cheaper operations.

There is no trade-off between efficiency and explainability in this grid. The configuration that changes the explanation at constant accuracy, the 2-bit membrane, is dominated on every axis by the 4-bit weight model: same accuracy, IoU 0.739 against 0.438, energy 0.016 against 1.108 times the parent, memory 512 against 1 176 KiB. The configuration to deploy is the **joint 4-bit model**: accuracy at parity (0.9835), explanation at the floor (IoU 0.714 against 0.704), estimated energy about 1/60 of the parent, and one eighth of its memory (190 KiB).

Arm	Bits	SOP (10^6)	Energy (µJ)	× FP32	Memory (KiB)
FP32	32	50.50	62.76	1.000	1520.2
Weights	8	51.02	2.28	0.037	655.7
Membrane	8	51.34	63.52	1.011	1244.6
Joint	8	52.29	2.32	0.037	380.0
Weights	4	54.21	1.00*	0.016	511.6
Membrane	4	52.62	64.67	1.031	1198.6
Joint	4	54.84	1.01*	**0.016**	**190.0**
Weights	2	74.60	0.61*	0.010	439.5
Membrane	2	57.81	69.34	1.108	1175.7
Joint	2	76.06	0.62*	0.010	95.0

Table 5.19: Synaptic operations, estimated energy and memory per inference, mean over three seeds, from the firing rates of each selected checkpoint. "× FP32" is the mean of the per-seed ratios to the parent. *Extrapolated per-operation costs. Memory is weights plus membrane state.

One assumption decides the size of the saving. The first convolution receives the analog image, and Table 5.19 costs it at the weight bit-width, which assumes the input is quantized to the same width. At 4 bits that layer is 19 % of the joint model's energy. If it is instead costed in floating point, the joint 4-bit saving factor falls from 62 to 3.5. The operation counts and the memory column do not depend on the assumption, and neither does the conclusion that the membrane arm is dominated; the factor by which the deployable model is cheaper does.

5.9.8 Still to be measured

TO DO. Two analyses need no GPU, the data being already on disk: the Earth Mover's Distance between the per-neuron firing-rate distributions of each model and its parent (from the `_rates.npz` files; compare with Smith et al. and with the seed-to-seed 0.039), and the disagreement subset (validation images correct in the parent and wrong after quantization, joined on the `index` column of the `_per_sample.csv` files and characterized by their physical descriptors). Add them here or drop the two mentions in Section 4.6 and Section 4.7.

5.10 Test-set evaluation

Model	Test accuracy	95 % Wilson interval
FP32 reference	[–]	[–]
Joint, 4 bit	[–]	[–]
Membrane, 2 bit	[–]	[–]

Table 5.20: Accuracy on the test partition (1 436 images), opened once.

TO DO. Run `evaluate_test.py` once, last, on the three models above (the reference, the recommended deployment configuration, and the configuration with the changed explanation). Add per-class recall.

Chapter 6
Discussion and Conclusions

6.1 The instrument

This thesis set out to measure whether a quantized spiking neural network reasons like its full-precision parent, and found that the measurement could not be interpreted until the instrument was calibrated.

A four-block convolutional spiking network classifies four families of transient noise in LIGO data at 0.9833 ± 0.0014 validation accuracy over three seeds, under constraints — shift-implementable membrane decay, at most eight time steps, 112-pixel input — that keep the configuration deployable by construction rather than by subsequent rounding. Each of those constraints was shown to cost nothing measurable in accuracy, and the resolution choice improved the health of the deepest layer by 12.7 percentage points of dead fraction.

The Spike Activation Map was implemented, its summation range corrected, and its free parameter fixed on the temporal axis it governs rather than on time-aggregated maps that cannot see it. Its faithfulness was tested quantitatively and found to hold on two classes of four, with a geometric explanation for the other two.

The calibration established three levels: the divergence produced by weight noise as large as an entire 8-bit quantization step (PCC 0.975), by a thirtyfold change in the explainer (PCC 0.960), and by initialization alone (PCC 0.153). The third is far larger than expected, and it carries three consequences.

Spatial metrics cannot compare independently trained models. Two networks that agree to within one validation sample in accuracy produce explanations correlating at 0.153. The mechanism is representational permutation rather than any defect of the metric, and the disagreement is uniform across samples rather than driven by outliers.

They can compare a model with its own parent. Under a perturbation 85 times smaller than the 2-bit quantization step, PCC remains above 0.975. The two results answer different questions, and the distinction must be made explicitly whenever such a number is reported. In this thesis it decided the design of the grid: every quantized model is fine-tuned from its own parent.

Published figures are not interpretable without these floors. Values around 0.85 are presented in the literature as evidence of preserved explanations. For this architecture that value falls between the numerical floor and the independent-solutions level, so it cannot be read without knowing which comparison produced it. Establishing the floors requires no training runs beyond a multi-seed baseline that is good practice regardless.

A fourth observation cuts across all of them: the class sensitivity ordering is identical under three unrelated interventions. Explanation sensitivity is a property of the class mor-

phology, not of the perturbation applied, and Extremely_Loud — saturated and spatially extensive — registers no change under any measure tested.

The preparation of the membrane arm added a result of its own: the network makes no use of negative membrane potentials. Clipping the hidden state to $[0, 2]$, which removes up to 47 thresholds of negative range, costs one validation sample and leaves the dynamics unchanged, so the membrane can be stored on an unsigned grid at one bit less than a symmetric one would need.

6.2 Quantization and explanation fidelity

Accuracy is not a sufficient criterion for spiking networks either. The finding of Rabbi et al. [33] on convolutional networks carries over, in a sharper form. Two configurations of this network classify equally well, a 4-bit weight model and a 2-bit membrane model, and their explanations differ by a factor of almost two in agreement with the original. Anyone selecting a configuration on accuracy alone could have chosen the second and deployed a model that reaches the right labels while showing a different part of the spectrogram. In the detector-characterization setting of Chapter 1, that is the scenario that matters: the label is correct, the diagnostic cue is not the one the original model would have given.

The divergence is real but bounded. The changed explanations are still faithful: deleting the pixels they highlight degrades the model's confidence almost as much as deleting the parent's highlights does, and the quantized model can be explained with its parent's maps as well as with its own. The quantized network has not stopped using spectrogram evidence; among several redundant sets of evidence it now highlights a different one. A faithfulness test alone would have declared the explanation intact; an overlap test alone would have suggested it was lost. Both are needed to say what happened.

The hypothesis was wrong in its form and right in its mechanism. The prediction that the membrane would move the temporal axis of the explanation before the weights moved its spatial axis is not supported: both arms first register at 4 bits, and at 2 bits the membrane changes where the explanation points more than when. What the membrane does change first is the temporal dynamics of the network itself: at 2 bits, with a single representable level between rest and threshold, the first layer fires with perfectly regular intervals. The temporal centre of mass, built to capture the time axis of the explanation, is blind to this, because it measures where the spike mass falls, not how it is spaced. A temporal explanation metric for spiking networks needs to see regularity as well as timing, and the present instrument does not.

The comparison with Smith et al. inverts. Smith et al. [38] report that the membrane needs at least four bits while learned weights survive two. The weight result agrees with theirs: uniform ternary weights fail, where they needed a learned quantizer to keep two bits. The membrane result does not: with a grid fitted to the measured range, a 2-bit membrane keeps accuracy at parity, where they lose 4–8 %.

Weak events are not where the explanation fails. The expectation that quantization would disturb spike decisions mostly in weak events, whose potentials sit near threshold, finds no consistent support: the gradient with event strength changes sign across seeds. The only consistent effect is reversed: the most saturated Extremely_Loud events lose their explanation most, because their evidence is spread over the whole image and the highlighted region is least constrained.

Efficiency and explainability do not trade off here. The configuration that changes the explanation at constant accuracy is the 2-bit membrane, and it is the one that saves no energy: with weights in full precision the per-operation cost stays the same while firing rates rise. The deployable configuration, joint 4-bit, keeps accuracy at parity and the explanation at the numerical floor, at about one sixtieth of the estimated energy and one eighth of the memory. The divergence between the three curves of Figure 5.10 exists, but it occurs at a point no one would choose for efficiency.

Seed variance is the dominant source of variation. Three full-precision networks of equal accuracy differ by a factor of 2.6 in the firing rate of their first layer, by more than half in estimated energy, and in which classes SAM can explain faithfully at all. The 2-bit membrane IoU ranges from 0.26 to 0.62 across seeds. Every comparison in this thesis was made within a seed, and a single-seed study of the same question could have reached any conclusion from "the explanation is preserved" to "it is lost".

6.3 Limitations

Generalization of the thresholds. The three calibration levels are specific to this architecture, this dataset and this explainer. The *mechanism* behind the collapse, representational permutation between independently trained networks, is general and well documented; the *magnitude* is not, and a different architecture could place the floors elsewhere. What generalizes is the claim that the floors must be measured, not the values measured here.

Labels are model output. Accuracy is agreement with the Gravity Spy convolutional classifier rather than with physical ground truth. A model that reproduced that classifier's systematic errors would score well here. The per-class confidences are available and would permit stratifying accuracy by the reference classifier's own certainty, which was not done.

The partition is random with respect to time. If glitches of the same class are correlated within observing periods, a random test set overestimates generalization to new periods. A block split by observing run would be the more conservative validation; the timestamps were dropped during feature selection for the reasons given in Chapter 3 and would have to be recovered.

T **sits at the imposed ceiling.** The search was forbidden from exploring more than eight time steps, so it is not known whether it would have preferred a longer integration window. An unconstrained search selected seven, which bounds the likely cost but does not eliminate the question.

The deletion metric fails on two classes. The geometric explanation offered in Section 5.5.2 — that deletion-based tests penalize spatially extended evidence — is an interpretation, not a demonstration. An insertion curve, which rewards rather than penalizes distributed evidence, would test it directly.

The explanation is class-agnostic. SAM does not use labels, so the deletion metric measures faithfulness to the model's *predicted* class. This is the correct reading of the metric and matches its original formulation, but it means misclassified examples are informative and should be inspected separately rather than pooled.

The temporal metric is partly kernel arithmetic. The γ analysis showed that the centre of mass of the SAM energy is dominated by a minority of near-continuously firing neurons. Between models compared at the same γ this component cancels, but it means that the CoM is most sensitive to changes in that minority and may under-report changes elsewhere.

Quantization is simulated. Fake quantization reproduces the values an integer implementation would hold, not its arithmetic: the membrane is quantized as a stored state after each spike decision, while on integer hardware the comparison with threshold would itself be performed on integers. The energy figures are analytical, at a 45 nm reference node, and below eight bits they rest on an extrapolation of the per-operation costs.

Sample sizes of the explanation analyses. Divergence is measured on twenty validation images per class, faithfulness on ten, stratification on fifteen per class and tercile, and the membrane range on a single checkpoint. These are sufficient for the effects reported here, which are large, but limit the resolution with which small differences between configurations can be ranked; the faithfulness analysis covers only the 2-bit membrane arm.

One grid, one fitting procedure. Each bit-width was fitted once per seed, with one fine-tuning schedule and one membrane range. The collapse of the 2-bit weight arm and the survival of the 2-bit membrane are properties of these quantizers; a different weight quantizer (learned scales, a wider grid for the readout) could move the weight arm's breaking point.

The control does not isolate the fine-tuning. Fine-tuning without a quantizer never improved the parent's validation accuracy, so the selected control checkpoint is the parent itself. The control therefore shows that the pipeline is inert, but not what the extra epochs alone would do to a model that does change; the quantized checkpoints combine both effects.

Checkpoints are selected on the validation partition on which they are then analyzed. With selection by accuracy and analysis of explanations, the dependence is weak, but the test-set evaluation is the only fully independent estimate.

The saving factor rests on an assumption. Costing the analog first convolution at the weight bit-width gives a joint 4-bit saving of 62; costing it in floating point gives 3.5. The operation counts, the memory and the ranking of configurations do not depend on it.

6.4 Future work

Deployment. Hardware deployment was deliberately left out of scope. The natural continuation is to export the selected integer configuration to a neuromorphic platform or a microcontroller, and to measure on it the latency, energy and memory that are estimated analytically here.

Normalization. The baseline contains no batch normalization, because quantizing its parameters is a problem distinct from quantizing weights and potentials. Deeper spiking networks need it, and both approximating it with shifts and removing it altogether through progressive training are available; repeating the study on a normalized architecture would test whether the conclusions survive.

The temporal axis of the input. Under direct encoding every temporal structure is generated by the network. Feeding the spectrogram column by column, so that time in the input corresponds to time in the signal, would give the temporal explanation a physical meaning and make the temporal metrics measure something a detector scientist can read.

Wider benchmark. Extending the four classes to the full Gravity Spy taxonomy, with a split by observing run, would make the classification results comparable with the published benchmark.

6.5 Closing remark

Compressed models are increasingly deployed where their explanations are acted upon. This thesis asked whether a quantized spiking network still decides for the same reasons as the network it came from, and found that the question has two prerequisites: knowing what the comparison metrics report when nothing has changed, and comparing every model with its own parent. With both in place, the answer for this network is that it mostly does — down to four bits, in every arm — and that the one configuration that does not is invisible to accuracy, partly invisible to correlation and to faithfulness, and visible to a metric that asks which pixels a reader is shown. That is the configuration a deployment chosen on accuracy could have picked, and the reason the question was worth asking.

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Appendix A
Additional figures

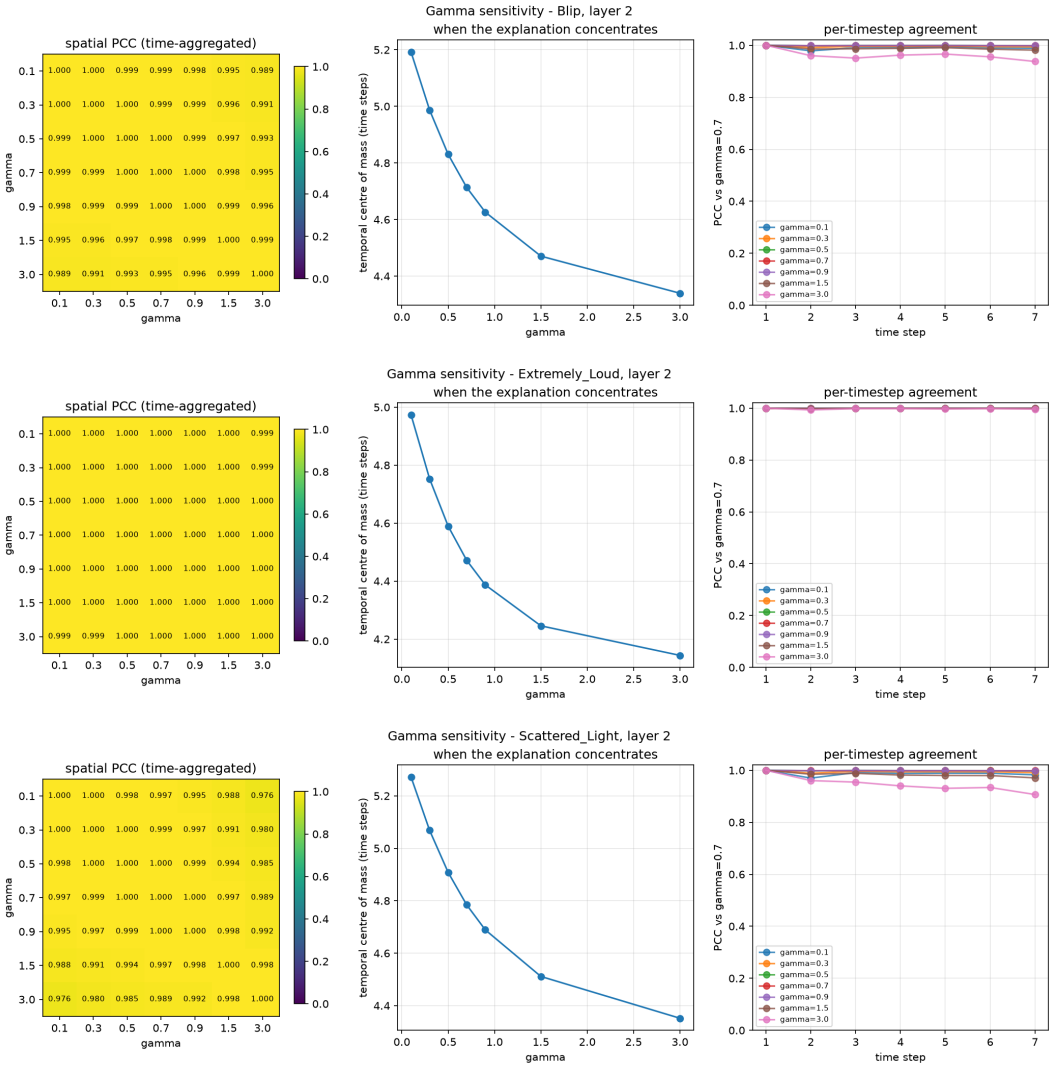


Figure A.1: γ sensitivity at layer 2 for Blip, Extremely_Loud and Scattered_Light (Violin_Mode is in Figure 5.8).

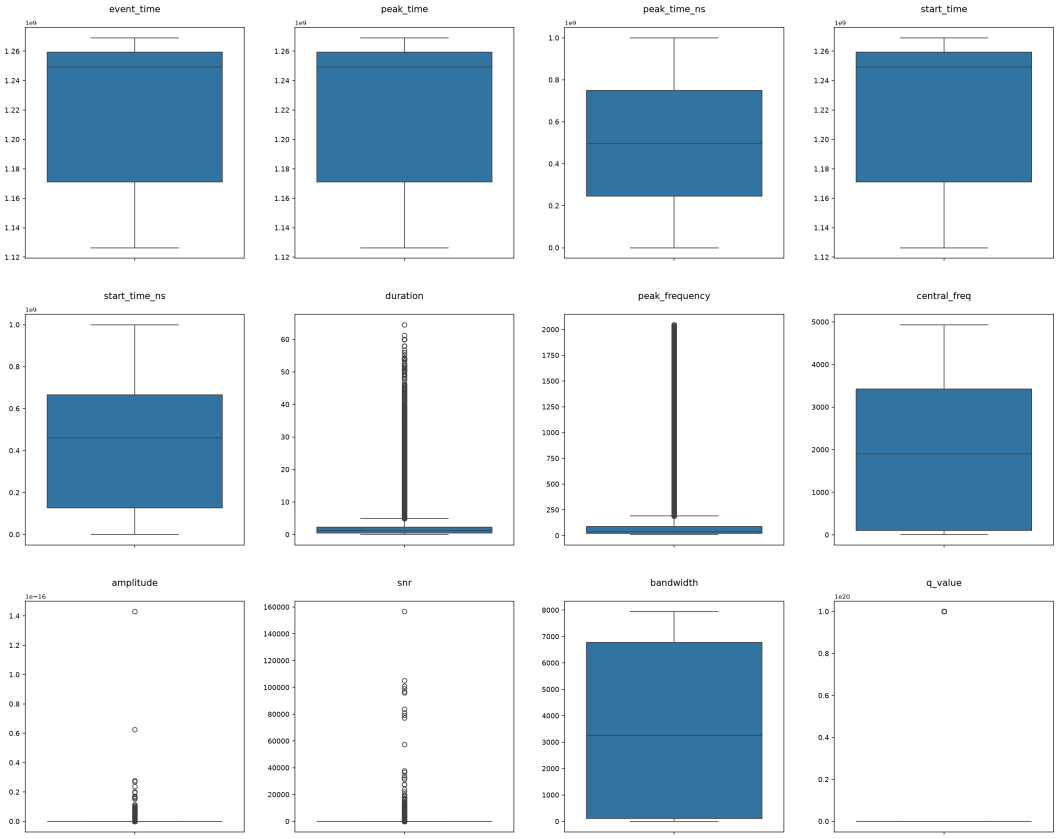


Figure A.2: Distributions of the twelve numerical Omicron descriptors over the full release. The quality factor collapses to a sentinel value of order 10^{20}; amplitude and SNR are dominated by a long tail of outliers.

Appendix B
Reproducibility

All code, configuration and result files are in the public repository

`https://github.com/devs-glitch/Tesi`

Table B.1 maps each result of this thesis to the script that produces it and the file it writes. Every script reads the model and the partition from `config/baseline_manifest.json`.

Result	Script	Output
Class selection, features	`scripts/data_prep.py`	`selected_dataset.csv`
Frozen partition	`scripts/freeze_split.py`	`split_assignment.csv`
Hyperparameter search	`scripts/tune.py`	`config/best_hyperparams.json`
Resolution study	`scripts/resolution_study.py`	`results/resolution_study.csv`
FP32 reference	`scripts/fp32_reference.py`	`config/baseline_manifest.json`
Dead neurons	`scripts/dead_neurons_check.py`	`results/dead_neurons.json`
Physical validation	`scripts/physical_validation.py`	figures
γ sensitivity	`xai/gamma_sensitivity_analysis.py`	`results/gamma_sensitivity.json`
Deletion metric	`xai/deletion_metric.py`	figure
Independent solutions	`xai/noise_floor.py`	`results/noise_floor.json`
Numerical floor	`xai/perturbation_floor.py`	`results/perturbation_floor.json`
Membrane range	`scripts/membrane_range.py`	`config/membrane.json`
Clipping study	`scripts/test_clipping.py`	`results/clipping_test.json`
Quantizers	`model/quantization.py`	—
PTQ gate	`scripts/ptq_check.py`	`results/ptq_check.json`
Fine-tuning grid	`scripts/grid_runner.py`, `train_quantised.py`	`results/grid_summary.json`
Per-model validation pass	`scripts/evaluate_model.py`	`results/grid/*.json, *.csv, *.npz`
SAM divergence	`xai/divergence_metrics.py`	`results/divergence.json`
Cross-faithfulness	`xai/cross_faithfulness.py`	`results/cross_faithfulness.json`
SNR stratification	`xai/stratify.py`	`results/stratified.json`
Energy and memory	`xai/energy.py -rates evaluated`	`results/energy.json`

Table B.1: From result to script.

TO DO. Add the scripts for the Earth Mover's Distance, the disagreement subset and the concluding figure once they are in the repository, and check every path against the final layout.